Retail Expansion in Developing Countries: Evidence from Hard Discount Stores

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August 24, 2026

Abstract

The expansion of modern retail in developing countries has reshaped an industry historically dominated by traditional, informal retailers. Its consequences for labor markets, however, remain poorly understood. We study the staggered entry of hard discount stores into small Colombian municipalities between 2010 and 2019. Combining social security records with household survey and tax data, we find that following the entry of a discount chain, the formal employment-to-population ratio increases by approximately 10 percent over time. Formality rises in retail, manufacturing, and construction, along with tax revenues in those sectors. We also find no significant effects on informal employment. Our results suggest that modern retail expansion in highly informal contexts can foster labor formalization.

Keywords: Hard discount stores, Modern retail, Competition, Employment, Formality.
JEL Codes: E24, J46, O17.

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1 Introduction

The expansion of modern retail in developing countries, which brings technological advances into sectors dominated by traditional outlets such as corner stores, mom-and-pop shops, and street stalls, has increased productivity (Lagakos, 2016), lowered prices and raised consumer welfare (Bronnenberg and Ellickson, 2015; Atkin et al., 2018), and reduced consumption at informal establishments (Bachas et al., 2024; Talamas Marcos, 2025). Yet little is known about its impact on formal and informal employment separately. This gap is important, as the formal–informal margin is central to policy in developing economies, where informality is widespread, costly to reduce, and has broad implications for tax collection, social protection, human capital, and aggregate productivity (Ulyssea, 2020; Bobba et al., 2022; Alfaro et al., 2025; Samaniego de la Parra and Fernandez Bujanda, 2024).

Modern retail has typically taken the form of Walmart supercenters, but this model has been largely absent in developing countries, as it requires large floor space and is usually located in low-cost land areas outside city centers. As Lagakos (2016) explain, this format also relies on widespread access to cars. Hard discount retailers changed this landscape. Smaller than regular supermarkets, they offer a limited product assortment, a high share of low-priced, own-label goods, and efficient logistics and operations (Jurgens, 2014; Sachon, 2010). This format enabled modern retail to expand into high-density residential neighborhoods and smaller municipalities that had been served almost exclusively by traditional retailers.

The hard discount model has become a major force in global retail, with sales exceeding 442 billion dollars in 2022 (Euromonitor International, 2023c,b; MarketLine Industry Profiles, 2023). Leading European chains, Aldi and Lidl, rank among the top 10 global retailers (Statista, 2023). This rapid expansion raises questions about its labor market effects. On the one hand, these stores generate formal jobs directly and indirectly through their suppliers. On the other hand, they intensify competition for incumbent retailers, as Bachas et al. (2024) document, which are highly informal, small-scale, and lack the organizational capacity and managerial strategies to respond to increased competition, potentially limiting the entry of informal retailers (Talamas Marcos, 2025).

In this paper, we examine the effect of a staggered expansion of hard discount stores (HDS) on formal and informal employment in a high-informality context. We use Colombia as a setting because, unlike in developed countries, over half of the workforce is employed informally, and 90% of businesses lack legal registration, making them informal as well (Otero-Cortés et al., 2025). Hard discount chains, however, are part of the modern retail sector, implying that they are formal firms that must hire workers formally, adhere to minimum wage laws, comply with tax laws, and maintain strong linkages with upstream suppliers that also operate in the formal sector (Bronnenberg and Ellickson, 2015).

Since the entry of the first hard discounter in 2009, the retail sector has undergone a substantial transformation. Through sizable investments, the leading chains have established more than 3,000 stores across more than 400 municipalities as of 2019 (Euromonitor International, 2023d). More efficient than supermarkets, with higher sales per square meter and lower operating costs, they rapidly expanded and overtook supermarkets by 2021, with major sales growth of 235% between 2017 and 2022 (Euromonitor International, 2023d). At the same time, social security records indicate that the three hard discount chains employed over 26,000 formal workers in 2019. Furthermore, discount store expansion can also cause formal employment spillovers onto other sectors, for instance, through upstream supply chain linkages (de Paula and Scheinkman, 2010), agglomeration forces (Evensen et al., 2023), and/or demand multiplier effects (Gerard et al., 2026).

To assess the impact of HDS, we assemble a unique dataset combining administrative records of social security contributions, which capture the universe of formal workers over time, subsidized social protection beneficiaries, labor force survey data, tax revenues, and information on each store's location and opening year for the three leading hard discount chains in Colombia from 2010 to 2019 at the municipal level.[1] Our empirical strategy focuses on small urban municipalities, in the spirit of Bresnahan and Reiss (1990, 1991), where modern retail entry induces a relatively large shock to local market structure given initial labor force size and high informality. Nonetheless, identifying the causal effect of HDS entry on local formal employment is challenging because

[1]Formal workers are defined as those who contribute to social security, and informal workers are those who do not comply with social security contributions.

store location decisions are endogenous. To address this issue, we leverage the rapid, staggered rollout of HDS across the country, based on logistical and market-size factors. Using an event-study research design, we quantify the dynamic effects of HDS entry on formal and informal employment in small urban municipalities, using the estimator of Callaway and Sant'Anna (2021).

Our identification strategy does not assume that store opening decisions are exogenous. In fact, we show differential trends between treated and never-treated municipalities before the arrival of the first discounter, which motivates using not-yet-treated municipalities as our control group. We claim that the *timing* of the first hard discount store opening, excluding those in capital cities, is unrelated to local unobserved trends. Therefore, by comparing cohorts of small urban municipalities where hard discount chains opened earlier with those where they opened later, we identify the effect of HDS on different outcomes.

To start, we provide suggestive evidence that the *timing* assumption holds in our scenario, as the treated and not-yet-treated groups exhibit similar trends in several economic outcomes, including employment, wages, and taxes, up to 5 years before HDS entry. A potential threat to our identification is the presence of time-varying shocks that are correlated with the *timing* of the arrival of hard discounters, such as spillover shocks from the main capital cities, which are potentially more pronounced in the early-treated municipalities, or additional firm entry at the same time as the hard discount chains. We address such potential concerns by controlling for distance in our baseline specifications and by showing that there is no immediate increase in the growth of formal firms.

We find three main results. First, the entry of a hard discount store in a municipality leads to a 10% increase in the formal employment-to-population ratio, as measured by both administrative and survey data. The growth is robust to violations of pre-trends and remains statistically significant when treatment timing is randomized (Rambachan and Roth, 2023; Roth and Sant'Anna, 2023). This result is primarily driven by increased formal employment in retail, manufacturing, and construction. We discuss the mechanisms underlying these inter-sectoral spillovers from retail, operating through both firm-level and consumer-level channels. On the firm side, HDS may stimulate local demand for intermediate inputs, benefiting complementary industries. In this context, discounters have incentives to source some goods locally, as transportation costs in Colombia are among the highest in the world (Londoño-Kent, 2009), and this aligns with their reliance on domestic suppliers (Asmar Soto, 2020). On the consumer side, hard discount chains are large formal firms that are more productive and have access to cheaper formal credit. They can therefore pay higher wages to attract both formal and informal workers (Ulyssea, 2020; Lagakos, 2016). By increasing workers' earnings, HDS entry may stimulate the local economy and generate multiplier effects (Gerard et al., 2026). While both channels likely operate, the evidence points to a larger role for the firm-side channel.

Second, we find a positive impact on local tax revenues, as the ratio of taxes to baseline public revenues increases by an average of 7.5% after the entry of discount stores. This increase is primarily driven by manufacturing and commerce taxes, supporting the idea that HDS have positive spillovers onto other sectors. Third, while our estimates for informal retail employment are small and sometimes positive, we cannot rule out that HDS reduce the entry of informal neighborhood shops due to increased competition, as we lack data to test this hypothesis directly. In this respect, Talamas Marcos (2025) show that an additional convenience store entry in a neighborhood adversely affects the creation of neighborhood shops in Mexico. The study also finds that incumbent neighborhood shops survive by leveraging their comparative advantage in supplying fresh products, offering lower prices due to their low fixed costs from avoiding tax compliance and operating from owners' homes (Ramos-Menchelli and Sverdlin-Lisker, 2023), and reducing costs by shrinking inventories.[2] Consistent with this, we also observe an imprecise negative effect on self-reported labor income of informal retailers in the later years of treatment. Taken together, our findings suggest that HDS can promote local economic formalization (through the labor market and consumption market) in developing countries.

[2]In Rwanda's coffee sector, Macchiavello and Morjaria (2021) show that increased competition weakens relational contracts, raising default and reducing both farmer welfare and mill profits.

Literature. This paper contributes to different strands of the literature. Most existing studies on the labor market effects of retail chain expansion focus on Walmart's expansion. Walmart's entry into US counties negatively impacts local labor markets by reducing retail employment and earnings in the mid- to long-run, as the company exploits its monopsony power and adversely affects other local retailers (Basker, 2005; Neumark et al., 2008; Wiltshire, 2026; Haltiwanger et al., 2010; Dube et al., 2007). Related literature explores the impact of expanding e-commerce fulfillment centers (FCs), such as Amazon's, in the US (Chava et al., 2023; Cunningham, 2025). Similar to our empirical strategy, Chava et al. (2023) exploit the staggered rollout of FCs, using areas not yet treated as controls, and find negative effects on retail employment but positive employment spillovers in transportation and warehousing, whereas Cunningham (2025) find negative effects in tradable services. Similarly, Greenstone et al. (2010) examine agglomeration spillovers from the arrival of "Million Dollar Plants" in the US by comparing counties where they enter with those that narrowly lose them.

Our paper differs from this literature in two key aspects. First, hard discount chains fundamentally differ from Walmart or Amazon FCs, as they act as a "bridge" between traditional and modern retail. HDS are located within walking distance of their customers, are smaller in size and in the number of workers they hire, and have a different business model. Second, our analysis context is different: we study the entry of modern, formal retail in a setting characterized by high informality, where informal competitors coexist with modern retailers. In this sense, our work relates to Bondi et al. (2025) and Busso and Galiani (2019) as we study the effects of the entry of modern retailers in settings in which traditional retail was highly prevalent, but our focus is on the effects on formal and informal employment and not on prices.[3]

To the best of our knowledge, this is the first paper to estimate the impact of HDS across the formal and informal labor markets in developing countries.[4] Our work is most related to Tala-

[3]There are papers studying Walmart's entry decisions in the US and its effect on incumbent retailers, such as Jia (2008), Holmes (2011) and Arcidiacono et al. (2020), but, as we explained, our work does not focus on prices or the effect on incumbent firms.

[4]In developed economies, Cho et al. (2015) study the impact of large discount stores in Korea, finding positive impacts on local retail employment driven by the large discounters and by the positive spillover effects on other retail sectors. Whereas Evensen et al. (2023) examine how the expansion of discounters affects incumbent local grocery

mas Marcos (2025), which analyze the impact of chain-run convenience store expansion in Mexico, yet our focus is not on the responses from neighborhood informal shops. Instead, we center on the implications for formal employment and local tax revenues. This paper also relates to Atkin et al. (2018), who analyze the effects on employment at the municipality level of Walmart's arrival in Mexico, finding no average effects, but in our context, the main difference is that HDS chains have strong linkages with national suppliers as their business model heavily relies on having their own low-cost brands that are made in Colombia, which is not the case for Walmart in Mexico. Therefore, we also aim to identify spillover effects of HDS entry on employment in other industries, such as manufacturing and construction, contributing to the literature on supply chain effects, as hard discounters have incentives to source their products from local formal suppliers (de Paula and Scheinkman, 2010; Gerard et al., 2023; Rios and Setharam, 2018).

This paper is structured as follows. The next section provides the institutional context of the retail sector in Colombia. Section 3 describes our primary data sources on local labor markets and how we measure the arrival of HDS in the municipalities. Section 4 discusses our identification strategy and its assumptions. Section 5 presents our results on employment, taxes, and labor income. We provide extensive robustness checks in Section 6. We conclude in Section 7.

2 Labor Market Context and Retail Sector

In Colombia, informal employment is the predominant form of employment. However, in recent years, there has been a noticeable upward trend in formal employment rates in both capital cities and other urban areas (see Appendix Figure F.1). Several factors contribute to this phenomenon. One key factor is the relatively lower labor costs, since the end of 2012, for employers to hire workers formally (Fernández and Villar, 2017; Morales and Medina, 2017; Kugler et al., 2017).[5]

stores in Norway. This paper identifies two opposing effects on sales and consumer traffic: a positive effect driven by store complementarity and a negative effect from fiercer competition. The agglomeration effect dominates when new discounters are located near existing retailers, while the competitive effect prevails when the distance between new and established stores increases.

[5]There are other events that, in turn, might have attenuated the increase in formal employment after 2015, like the arrival of Venezuelan immigrants (Delgado-Prieto, 2024a).

In this context, we want to study the impact of the arrival of HDS on local formal employment, as they only hire workers formally and demand inputs only from legally registered suppliers, and determine if this contributes to the observed national trends.[6]

2.1 Retail Sector

The grocery retail market in Colombia is a sizable 40 billion-dollar market (Euromonitor International, 2023d). It represents around 13% of South America's retail market and, at the national level, accounts for more than half of total retail sales (MarketLine, 2023). Three actors have historically played a significant role in this market: small, traditional neighborhood shops (mainly informal), supermarkets, and hypermarkets. According to market data from Euromonitor International (2023d), in 2017 neighborhood shops accounted for 52% of retail grocery sales, while modern retailers accounted for around 23% of the sector's income.

Neighborhood shops are mostly informal businesses that offer a limited supply of essential goods, primarily in residential areas.[7] They tend to create a close relationship with their customers, even offering informal credit to them (Talamas Marcos, 2025).[8] Neighborhood shop prices are not necessarily lower than supermarket prices, yet as they sell smaller quantities of essential items, the ticket for daily grocery shopping would fit into lower-income households' daily budgets. In contrast, supermarkets are large, formal retailers with infrastructure similar to that in the US, where customers can find goods ranging from fresh produce to home appliances. In this context, the arrival of HDS increases competition in the market for certain products among supermarkets and neighborhood shops (Sánchez Duarte, 2017). We provide further descriptive statistics on profits, employment, and the formality of neighborhood shops, using data from the Micro Businesses

[6] The 2012 Colombian tax reform could potentially confound the impacts from the arrival of HDS. However, this reform was enacted at the national level in 2012 and applied to all firms with at least 2 employees that earned up to 10 times the minimum monthly wage. Thus, it is unlikely that there is a correlation between the first arrival of a hard discount chain in a municipality and the treatment intensity of the 2012 reform.

[7] Neighborhood shops are not to be confused with convenience stores such as *Oxxo* or *7-Eleven*.

[8] Neighborhood shops are prevalent in the country. In 2019, there were approximately 270,000 stores in this category across the 100 largest municipalities in Colombia (Fenalco, 2019), with median weekly sales of around 350 USD and an average ticket per customer of 1.50 USD. They target low-income households that typically buy only everyday groceries.

Survey, as reported in Appendix A.

The supermarkets and hypermarkets segment, an important source of formal employment, has been largely dominated by three firms: *Grupo Éxito* (with the *Éxito* and *Carulla* brands), *Supertiendas & Droguerías Olímpica* (*Olímpica* and *Sao*), and *Cencosud S.A.* (*Jumbo* and *Metro*). This format was introduced in Colombia in the late 1940s by José Carulla Vidal (Silva Guerra, 2011), inspired by Mexico's growing supermarket chains like *Sumesa*. During the 1950s, supermarkets spread to major Colombian cities with new owners and chains (Redacción Semana, 2015). According to Euromonitor International (2023d), the sector has a market size of about 4 billion dollars, with around 2,200 outlets. The average store size is 750 square meters, with annual sales of US$ 2 million (US$2,450 per square meter).

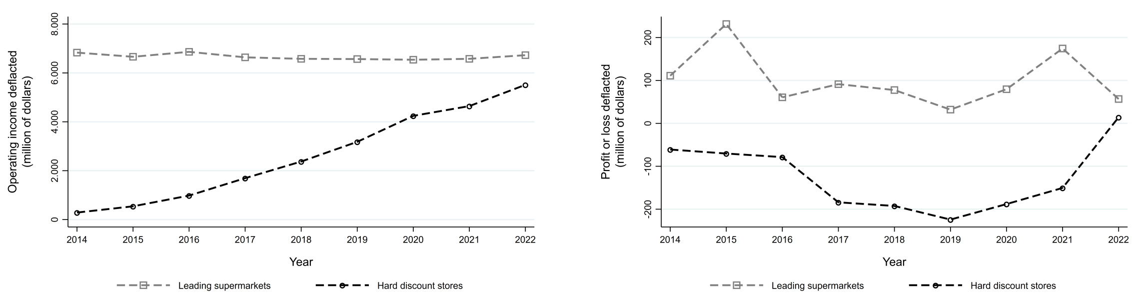
By the late 1990s, hypermarkets had entered the market with a clear socioeconomic segmentation of customers. Some chains offered lower prices to target low- to mid-income consumers, not through the hard-discount format, but by offering lower-quality products. Colombia has about 210 hypermarkets, mostly in large cities, generating 4 billion dollars in sales (Euromonitor International, 2023d). The average hypermarket has 4,700 square meters of selling space and generates around US$ 19 million in annual sales (US$4,020 per square meter).

Following the model of European hard discounters, such as Aldi, the first hard discount chain (*D1*) opened in Colombia in 2009. The second, *Ara* (owned by the Portuguese retail group Jerónimo Martins), followed in 2012, and the third, *Justo & Bueno*, in 2016. HDS stores are smaller than regular supermarkets, typically 250 to 300 square meters, and achieve lower operating costs through strategies such as streamlined distribution with a limited product range, minimal advertising, displaying goods in shipping boxes, and lean staffing. HDS have expanded into residential neighborhoods and city centers, using a low-cost business strategy that acts as a bridge between modern and traditional retail (Bachas et al., 2024; Lagakos, 2016). Though in 2019, HDS had lower annual sales than supermarkets, they generated around US$3,472 per square meter, 41% more than traditional supermarkets (Euromonitor International, 2023d).[9]

[9]Still, hard discounters offer a more limited product range, which means they are not perfect substitutes for supermarkets and can lead consumers to shop at multiple establishments. For instance, Florez-Acosta and Herrera-

Appendix Figure C.1 shows the expansion of HDS from 2010 to 2019, both overall and by chain. Over the decade, they expanded rapidly across Colombia. The three main chains opened nearly 3,000 stores and employed around 26,000 workers across 408 of Colombia's 1,120 municipalities, with the remaining municipalities largely rural, particularly in the Amazonía and Orinoquía regions. Figure 1 compares operating income and profits between supermarkets and hard discounters. Supermarkets have historically led in both measures, but the gap in operating income has narrowed substantially. More recently, the leading hard discounter has surpassed the top supermarket in operating income. In terms of profits, the two main hard discounters (the third exited in 2021) only became profitable in 2022, after years of investment. In contrast, the three major supermarkets did not experience a decline in profits until that same year.[10]

Figure 1: Operating income and profits of the leading supermarkets and hard discounters



(a) Operating income (b) Profits

Note: We aggregate the operating income and profits of the three leading supermarkets and the three leading hard discounters in the country, expressed in real terms (CPI 2018=100). We use the exchange rate from December 2018: 1 USD = 3,250 COP. Source: Supersociedades (Operating income and Profits), Banco de la República (Exchange rate), and DANE (CPI series).

The expansion of hard discount chains not only increased retail options for consumers but also brought substantial investment shocks to the municipalities where they opened. For example, in a 2018 interview, Ara's manager reported that opening 500 stores involved over 400 million euros,

Araujo (2020) report that French households commonly visit multiple supermarkets per week, even when they offer similar products, suggesting that HDS entry may accelerate multi-stop shopping behavior among consumers.

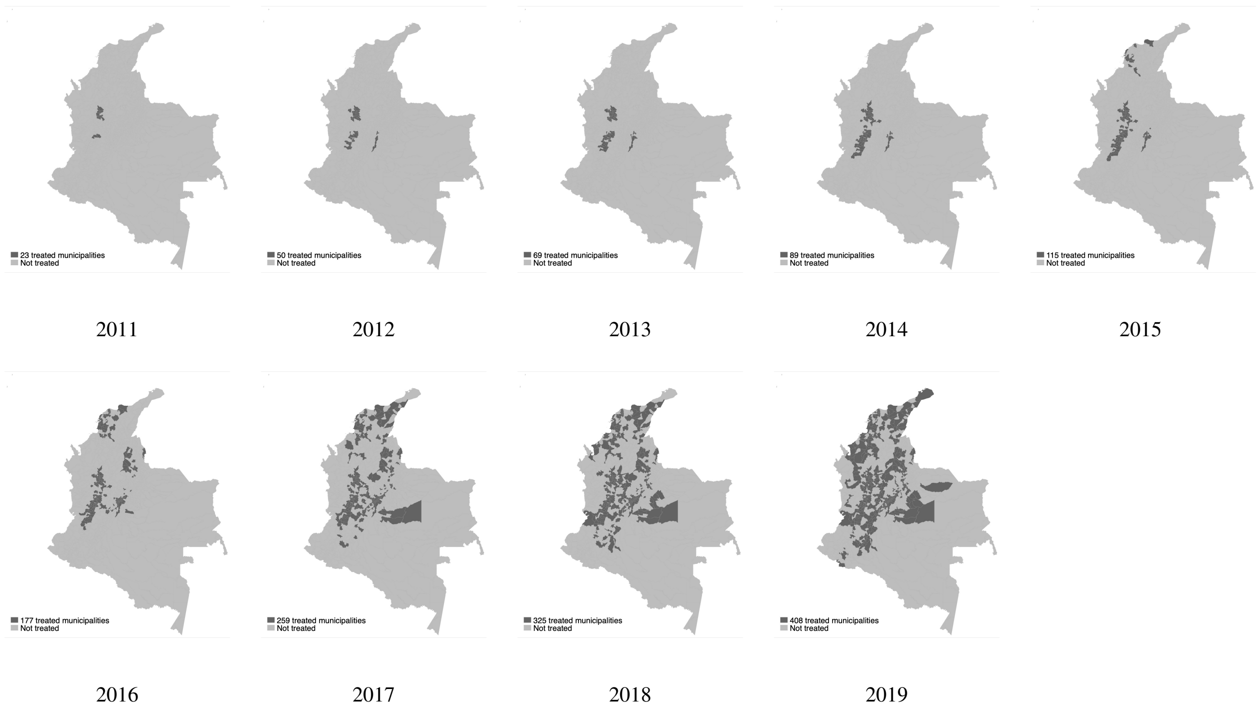
[10]In Appendix D, we examine whether hard discount chains differ from other formal firms in terms of wage premiums by estimating an AKM model. After controlling for worker characteristics, hard discount chains consistently pay higher premiums than the largest supermarket and most formal firms in the country.

about 0.9 million USD per store (Acosta, 2018; Morante, 2018). Similarly, in 2020, D1's manager stated the firm had invested 123 million USD to open 800 stores (roughly 154,000 USD per store, González Bell (2020)). As a benchmark, Ara's per-store investments are 1.9 times the median annual local tax revenue in our sample of municipalities, whereas for D1 they are 36%. In terms of investment expenditures, they represent 19% and 4%, respectively.[11] Many municipalities host multiple hard discount chains, greatly increasing the total investment received.

Figure 2 shows that hard discount store expansion began in Colombia's central region, then moved to the Caribbean and southern areas in a staggered pattern, as chains focused on different regional markets to grow aggressively. According to market data, by 2017 (when discounters already had over 1,700 outlets), hard discounters' sales grew by 235% between 2017 and 2022, compared with 29% in the aggregate grocery retail sector. The number of stores rose by 133%, and the selling space by 141%. The market share increased from 5.2% to 13%, surpassing hypermarkets in 2020 and supermarkets in 2021. By the end of 2022, there were more than 4,000 outlets (Euromonitor International, 2023d), with large-scale investment plans to increase the number of HDS in the following years (Almario, 2023), underscoring the policy relevance of studying their impacts on local labor markets.

[11] Median annual tax revenue in our sample (2010–2019) is 425,674 USD, and median investment expenditure is 4,150,126 USD, based on the Colombian municipalities panel (Acevedo and Bornacelly Olivella, 2014). We confirm the chains' investment figures using financial data from *Supersociedades*. Net cash flow from investing activities per new store was about 1.3 million USD for Ara, 47,000 USD for D1, and 90,000 USD for Justo & Bueno.

Figure 2: Geographic location of HDS



Note: This figure shows the geographic expansion of Colombia's three main hard discount chains, based on store stock data from 2011 to 2019. Source: Authors' calculations using public location data from hard discounters' websites.

In line with the geographical expansion, we show in Appendix C.5 that hard discount chains decide entry based on constant municipality characteristics, particularly population size and driving time to Bogotá and Medellín, rather than on economic development (measured through the rurality index) or sectoral composition (measured through the formal tertiary employment share). Among treated municipalities, the timing of entry is again best predicted by population and driving time to Bogotá, while economic development and other time-invariant characteristics add less explanatory power.

A potential concern with our *timing* assumption is that hard discount chains might instead decide when to open their stores based on local employment dynamics or tax collection, which would violate the parallel trends assumption underlying our design. We find no evidence of this after running cohort-specific regressions of the probability of store entry on the annual growth of formal employment, wages, and tax revenues, which show no significant relationship with entry timing (see Appendix C.5). The evidence therefore indicates that discounters based their decisions on fixed characteristics, such as potential market size and logistical convenience, rather than on the dynamics of municipal formality or tax collection, supporting our research design.

3 Data

For our treatment exposure, we measure the year a hard discount chain arrived in a municipality. We rely on web-scraped information on the universe of HDS for the three leading chains in 2019. This allows us to identify their location and, using business registration data from the Chambers of Commerce, determine the date of entry into a given municipality. Colombian law requires all firms to register their establishments, including stores and distribution centers. We use the store's registration date as a proxy for its opening date and match the web-scraped spatial data with the Chambers of Commerce data using the store's name and parent company.

Our final data set comprises 2,847 stores, their municipalities, and a proxy for the opening date. We then compute the year the first HDS opened in a municipality, yielding 414 observations (372 excluding department capital cities and municipalities where HDS arrived before 2011 or after 2019). To address potential measurement error in opening dates, we validate our entry data against an independent source from the largest supermarket chain in the country and find that, in the vast majority of cases, the opening year matches exactly or differs by at most 1 year at both the municipality and store levels. Appendix C explains the matching process in detail.

This paper uses multiple data sources to capture both formal and informal outcomes. First, we use the matched employer-employee data (PILA, by its acronym in Spanish) from 2010 to 2018. PILA contains the universe of mandatory social security contributions made on behalf of each formal worker in Colombia. These data allow us to identify formal employment by industry level and per municipality.

For the main analysis, we exclude capital cities, where the relative market shock from store openings is small, and timing may be influenced by labor market trends, thereby violating our identification assumption (we present evidence of this below).[12] We also exclude municipalities without HDS by 2019, as their labor markets (typically small or rural) likely follow different dynamics. The final estimation sample focuses on small urban municipalities, with an average of 1.6 million observations per month across 372 of the country's 1,120 municipalities, covering 38% of the population. To address limitations in identifying municipalities and industries for some firms in certain years, we use the panel component of the data to extract information from other years.[13] Still, industry-level results may be noisier because of misclassifications by firms in all years.[14]

As the main source of informality, we use a monthly cross-sectional household survey that covers approximately 240,000 households per year (GEIH, by its acronym in Spanish). GEIH is Colombia's labor force survey and has extensive sample coverage across the country, though not in all municipalities where hard discounters are present. Thus, the estimation sample of municipalities using GEIH reduces to 191 from the 372 municipalities we observe in PILA. However, GEIH allows us to characterize informal and formal labor outcomes by using survey questions on workers' contributions to the social security system to categorize workers as informal or formal. For the analysis, we aggregate this information at the municipality level using department survey weights after restricting to individuals in urban areas between 18 and 64 years, and for the wage analysis, we further consider only workers with positive labor income.[15]

The GEIH survey is not representative at the municipal level, even though the municipalities included are among the largest surveyed outside capital cities. However, since our analysis aggre-

[12]This restriction removes around 80% of observations from the PILA, as most formal jobs are concentrated in capital cities.

[13]We have missing municipality information for some observations from 2017 to 2019. We impute it using the mode of the municipalities in which the same worker was previously registered or, when this is also missing, the worker's municipality in 2020. After this procedure, we exclude 1.8% of PILA workers from our sample because of an incorrect municipality code.

[14]Industry classification in PILA is self-reported by firms using a four-digit International Standard Industrial Classification (ISIC) code. ISIC Revision 3 was the standard until 2013, after which the classification was updated to Revision 4. From 2013 to 2019, some firms continued to report codes from the old list. We therefore compute each firm's modal four-digit code and classify it using Revision 4 when possible and Revision 3 otherwise. Approximately 1% of workers per month in our estimation sample cannot be classified to an industry because of incorrect ISIC codes.

[15]Because the municipality information of the GEIH survey respondents is not publicly available, we obtained it through specialized DANE data centers.

gates similar municipalities into treatment cohorts ranging from 5 to 35 municipalities (Table 2), our identifying variation occurs at the cohort, not municipal level, mitigating concerns about noise due to lack of statistical representativeness. We also aggregate monthly surveys into annual data, further reducing noise. Notably, our GEIH-based estimates of formal employment align with those of administrative data, supporting the survey's reliability. Appendix Figure E.1 further shows that re-estimating the administrative-data results in the municipalities covered by GEIH yields similar, albeit noisier, estimates. This comparison suggests that differences between the administrative and survey estimates primarily reflect treatment-effect heterogeneity across the two municipality samples rather than differences in data quality.

We also use the census of beneficiaries of subsidized social protection (SISBEN, by its acronym in Spanish) to approximate the stock of non-formal workers at the municipal level. Although this source covers the universe of low-income individuals in the country, including those outside the labor force or unemployed, it also includes informal workers who receive social protection subsidies. Therefore, SISBEN is a proxy we use to measure the size of informality at the local level.

To capture broader tax effects on the local economy, we use data from *Operaciones Efectivas de Caja*, collected by *Departamento Nacional de Planeación* (DNP, by its acronym in Spanish), which contains detailed information on revenue by each type of tax collected at the municipal level, such as industry and commerce tax, property tax, among others.

3.1 Descriptive Statistics

Administrative records (PILA). Formal employment in the services sector grew steadily from 2010 to 2019, driven by favorable economic conditions and labor reforms (Appendix Figure F.2).[16] Nonetheless, informality remains widespread, with about half of Colombia's workforce lacking

[16]Since the end of 2012, a reduction in the cost of hiring workers formally (Fernández and Villar, 2017; Morales and Medina, 2017; Kugler et al., 2017) increased incentives to formalize. Colombian legislation also made part-time formal employment relatively costlier than full-time, although regulatory changes in 2014 allowed weekly contracts that promoted part-time employment (de la Parra et al., 2024).

access to the pension system, and even higher rates outside capital cities, the focus of our analysis (Appendix Figure F.1). Amid persistent informality, the rise of hard discount chains represents a notable push toward formal employment. From employing fewer than 500 formal workers in 2012, these chains grew to nearly 26,000 employees by 2019 (Appendix Figure F.2 panel (b)), 86% of them full-time, averaging 10 workers per store by the end of the period.

To better understand who works in these chains, we analyze individual employment histories. Between 2010 and 2019, 57,963 individuals were employed by one of the three leading HDS. Of these, 16% entered formal employment for the first time, while 31% transitioned from previous formal jobs. First-time formal workers are younger (average age 24), 55% are women, and 47% start on part-time contracts. In contrast, job switchers are older (average age 30), 60% are men, and more than 75% come from retail, wholesale, or services. These patterns suggest that while retail has traditionally been an entry point into informal employment, discount stores expand formal opportunities within the sector, particularly for younger workers.

Appendix Table F.1 presents descriptive statistics for the municipalities in our estimation sample. Treated municipalities in 2011 had an average of around 7,000 formal workers, a number that declined as HDS expanded into smaller municipalities over time, up to around 4,000 workers in municipalities treated in 2018.

Labor force survey (GEIH). Table 1 shows the mean and standard deviation of several labor market outcomes by year and treatment status in our estimation sample.[17] The average of total employment by treatment cohorts, weighted by the employed population in 2010, shows that hard discount chains prioritized large municipalities for their initial openings (the employed population in the typical early-treated municipality was almost twice that of the typical not-treated-yet municipality in 2011). However, both numbers decreased over time, suggesting that they later opened in smaller municipalities.

Despite differences in employment between treated and untreated municipalities, there are no

[17] Appendix Table F.3 further shows descriptive statistics on wages and working hours for the formal and informal sectors.

significant differences between the two groups in most outcomes during the initial years of the expansion. Employment and inactivity rates were similar in 2013, as were wages and working hours, even when disaggregated by informality status.

The most significant differences stem from differences in informality and industry composition across municipalities. For instance, formal employment in 2013 represented, on average, 47% of total employment in treated municipalities, while the share was 37% in the not-yet-treated group (nearly a 9 pp gap). At the same time, the typical municipality with HDS presence by 2013 was less dependent on retail and more dependent on the primary and secondary sectors: retail workers represented 16% of 2010 total employment in the treated group, compared to 19% in the not-yet-treated group, and the respective shares for the primary and secondary sectors are 34% and 30%. Conversely, the rest of the commerce and services sectors had a similar weight in the local economies of both treated and untreated municipalities.

Table 1: Employment statistics for the estimation sample using survey data

	Treated				**Not yet treated**			
	2011	**2013**	**2016**	**2018**	**2011**	**2013**	**2016**	**2018**
Employment rate	68.6	70.4	72.3	70.8	69.7	71.1	68.7	68.1
	(1.9)	(6.4)	(5.1)	(6.4)	(7.0)	(6.0)	(7.3)	(9.3)
Unemployment rate	13.1	11.6	10.2	11.3	12.0	10.4	12.5	11.1
	(2.6)	(3.8)	(2.9)	(4.6)	(4.4)	(4.1)	(5.0)	(5.6)
Inactivity rate	21.0	20.7	19.5	20.2	21.0	20.7	21.6	23.7
	(2.9)	(5.4)	(5.0)	(6.1)	(6.6)	(5.6)	(6.0)	(7.9)
Employment share: Retail	19.2	16.2	18.8	19.5	18.1	19.4	21.1	17.4
	(3.3)	(4.2)	(5.0)	(6.6)	(6.6)	(6.4)	(8.4)	(9.4)
Employment share: CHR without retail	13.5	13.8	15.9	15.6	12.2	14.7	17.8	14.4
	(3.5)	(4.5)	(4.6)	(5.2)	(4.3)	(5.7)	(7.6)	(6.8)
Employment share: Primary and secondary	29.3	34.3	33.6	34.3	32.2	30.3	30.9	38.2
	(2.7)	(10.5)	(11.7)	(13.9)	(10.5)	(11.4)	(13.9)	(21.2)
Employment share: Services without commerce	39.1	43.8	48.9	49.4	42.5	46.7	46.5	45.1
	(2.4)	(10.1)	(10.5)	(12.2)	(8.9)	(10.3)	(13.8)	(18.5)
Informal employment share	56.3	61.0	65.1	70.0	70.1	73.4	78.8	80.0
	(11.6)	(15.6)	(17.6)	(21.1)	(17.1)	(20.5)	(22.2)	(36.5)
Formal employment share	44.8	47.1	52.0	48.6	34.8	37.7	37.3	33.6
	(12.5)	(13.6)	(16.1)	(17.9)	(14.5)	(13.7)	(15.0)	(13.2)
Municipalities	5	28	85	156	186	163	106	35
Average 2010 Employed Population	35,923	21,407	26,963	21,058	18,750	18,821	12,974	10,917

Note: This table reports the mean of selected labor market indicators using the municipal panel of the GEIH by year and treatment status. The employment, unemployment, and inactivity rates are constructed by dividing the number of employed, unemployed, and inactive individuals in a municipality over the 2010 municipal working-age population. For shares (by informality status or economic activity), we divided the number of workers in the sector by the total municipal employment in 2010. We use this fixed aggregate to weigh the mean and standard deviation. Standard deviations are reported in parentheses.

4 Empirical Specification: Cohort Analysis

For our identification strategy, we exploit the staggered rollout of HDS in Colombia to quantify its effects on local labor markets, similar to the empirical strategy used by Chava et al. (2023) and Cunningham (2025) to study the local impacts of Amazon FCs. Table 2 illustrates the staggered introduction of hard discount chains within our sample of municipalities, comprising 372 in PILA and 191 in GEIH. There are no municipalities in which the first hard discount store to open subsequently closes, so all municipalities in our sample remain treated over time.

Table 2: Cohorts of Treatment

	Municipalities (Admin)	Share	Municipalities (Survey)	Share
2011	10	2.7	5	2.6
2012	24	6.5	14	7.3
2013	19	5.1	9	4.7
2014	19	5.1	11	5.8
2015	23	6.2	12	6.3
2016	56	15.1	34	17.8
2017	76	20.4	45	23.6
2018	64	17.2	26	13.6
2019	81	21.8	35	18.3
Total	372	100.0	191	100.0

Note: In this table, we exclude the municipalities treated in 2009, 2010, and 2020 due to the small number of treated cities. Moreover, we dropped 24 capital cities from PILA and 23 from GEIH, plus an additional city from GEIH that did not appear in all survey years.

Using the canonical Two-Way Fixed Effects (TWFE) regression to estimate the Average Treatment Effect on the Treated (ATT) in these settings is common. Yet recent literature indicates that this estimation strategy may be biased because treatment effects are heterogeneous across treated cohorts (Goodman-Bacon, 2021; Sun and Abraham, 2021). For instance, treatment effects on earlier cohorts might differ from those on later cohorts, but the TWFE regression aggregates them into a single parameter using weights that can be hard to interpret or incorrect, leading to biased coefficients. Therefore, we use the differences-in-differences estimator of Callaway and Sant'Anna (2021) (C&S, hereafter), as it addresses the main issues of differential treatment timing and heterogeneity in treatment effects across cohorts.

The intuition behind the C&S estimator is that it allows us to break down all possible comparisons into a two-period, two-group (2x2) framework to estimate multiple ATT(g,t) for units treated in the same year (cohort g) and measured in period t. In our setup, cohorts are defined as the year in which the first hard discount store enters municipality l, so treatment starts with this first HDS entry. Therefore, if the treatment started in 2015, then $g = 2015$. Due to the small number of treated units at the beginning of treatment, we restrict the leads to $t-g < -5$ and the lags to $t-g > 5$. More concretely, they are estimated as follows:

$$ATT(g,t) = E(y_{l,t} - y_{l,g-1} \mid G_l = g) - E(y_{l,t} - y_{l,g-1} \mid G_l > t), \text{ for all } t \geq g. \tag{1}$$

Here, we use the not-yet-treated group ($G_l > t$) as a control group for a cleaner comparison with treated municipalities, estimate the differences relative to the baseline period $(g-1)$ using ordinary least squares without covariates, and test the robustness of these estimates to the inclusion of baseline controls. Then we aggregate all differences into an overall ATT using weights $w_{g,t}$ that are based on the number of treated units used in the particular ATT(g,t)[18]:

$$ATT_{post} = \sum_{t}^{T} \sum_{g}^{G} 1\{t \geq g\} w_{g,t} ATT(g,t). \tag{2}$$

Next, as a benchmark for the C&S coefficients in the event-study figures, we estimate a traditional TWFE regression for specific outcomes.[19] The model takes the following form:

$$y_{lt} = \gamma_l + \gamma_t + \sum_{k=-5, k\neq -1}^{5} \beta_k * \mathbf{1}\{k = t-g\} + \varepsilon_{lt}. \tag{3}$$

Here, the year that the first store opens equals g, and the years are denoted as t, so the relative event time indicators are k. We add time fixed effects (γ_t) and municipality fixed effects (γ_l) to the specification to control for unobserved constant characteristics over time for all units and

[18]For the overall ATT, we use eight cohorts $G \in \{2011, \ldots, 2018\}$ of treatment in the administrative and survey data.

[19]To compare the pre-treatment coefficients of TWFE and C&S accurately, we quantify long gaps when using C&S.

in all years for each unit, respectively. The parameters of interest are β_k, which come from k event-time dummy variables. These dynamic treatment effects measure the effect on y relative to an omitted period, corresponding to $k = -1$. In this specification, identification arises from two control groups: units that have not yet been treated or units that have never been treated (Schmidheiny and Siegloch, 2023). Again, we select municipalities that have not yet been treated as a control group.[20]

Before discussing the identifying assumptions, it is helpful to describe the functional form of the main outcomes. First, using PILA, we quantify the average real monthly wages or labor income in each l, after restricting to full-time workers and with 30 days of employment, and apply the standard log transformation. From GEIH, we also quantify average real monthly wages (labor income) for positive earners, as well as average weekly working hours, across the informal and formal sectors in each l, and then apply a log transformation. For the employment outcomes, we define them in employment-to-population ratios following other influential studies, such as Autor et al. (2013) and Dustmann et al. (2017), instead of logs given the presence of zeros (Chen and Roth, 2024) and the considerable variation in the distribution of treated and not-yet-treated outcomes in the baseline year (McConnell, 2023). Thus, we define a common denominator across our datasets for comparability: the working-age population in each l obtained from the 2005 population census, and, in the numerator, we use the number of individuals in l from PILA, SISBEN, or GEIH. In this way, we measure the relative growth rates of employment between treated and control municipalities, thereby avoiding the approximation biases inherent in growth rates based on log differences. We also use employment levels to capture the number of formal jobs created and consider log specifications as a robustness check, yielding results consistent with those based on ratios. For the tax outcomes, we define them as ratios, using the pre-treatment measure of public revenues in the denominator and, in the numerator, the specific revenue type that varies over time.

Defining employment variables as logs, levels, or ratios with a fixed denominator (the 2005 population) can be problematic if the population was growing faster in municipalities where stores

[20] To achieve identification using not-yet-treated municipalities as a control group, we bin or accumulate the leads as $k \leq -5$ and the lags as $k \geq 5$ (Borusyak et al., 2024).

arrived earlier than in the later-treated group. We rule out this concern by showing that the early arrival is not correlated with higher population growth. Appendix Table F.2 reports estimates of these correlations. We regressed municipal population growth rates on a dummy variable indicating whether hard-discount chains had arrived in 2013 or earlier. We used both pre-treatment population growth (how much the population increased from 2005 to 2010) and post-treatment (from 2010 to 2020) as dependent variables, and different samples based on dataset inclusion (i.e., whether the municipality is in the survey data). The results show that municipalities treated early did not experience higher population growth, neither in the pre-HDS period (before 2010) nor in the post-HDS period.

Identification assumptions. The main identifying assumption required is the unconditional parallel trends assumption (PTA). It establishes that the treated and control units would have evolved similarly in the absence of store openings. Because our control group comprises municipalities that have not yet been treated, we do not assert that store openings are exogenous to local economic trends. Instead, we argue that unobserved trends do not determine the *timing* of store openings. The rapid rollout of HDS across the country makes it less likely that local economic trends among the early- and late-treated cohorts determine the timing. Specifically, the *timing* of openings is largely determined by factors such as population size (larger cities are prioritized) and the municipality's proximity to the largest cities (see Appendix Table C.1). Our specification already absorbs any constant observed and unobserved characteristics associated with these factors. As a placebo test, we show there are no differential employment, wage, or tax trends before the arrival of these stores across different sectors and industries. This evidence supports our *timing* assumption for identifying the effects of HDS openings in local labor markets.

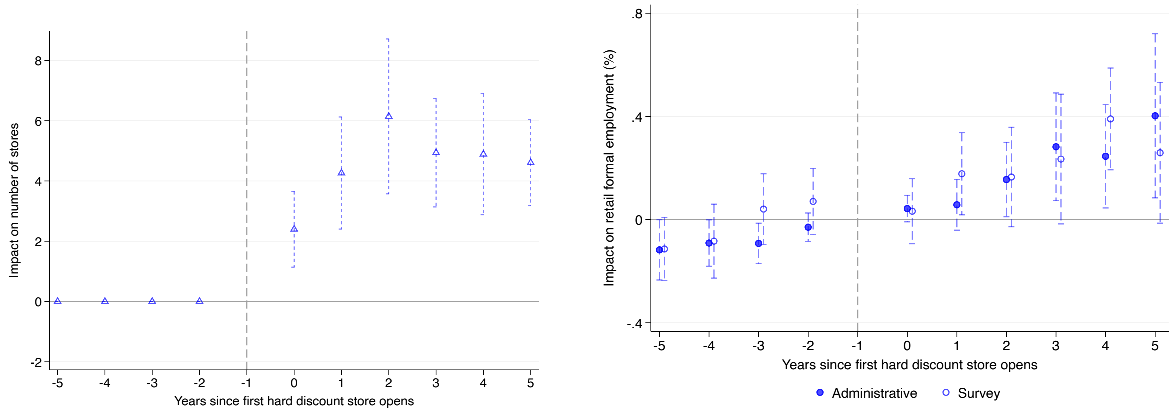
Our identification assumption could be affected by other formal firms entering municipalities simultaneously with hard discounters, spillover effects from capital cities to early-treated municipalities, and differences in the treatment's intensive margin across municipalities. We discuss these threats and present some reassuring evidence against them in Appendix Section B. We also argue for using the not-yet-treated group as our control group (rather than the never-treated group) and for excluding department capitals from our identification sample.

5 Effects on Employment, Taxes and Labor Income

We begin by presenting the direct effects of HDS entry. Panel (a) of Figure 3 documents the expansion margin, showing the average number of additional stores that open following the first entry. We find that the average number of stores in our estimation window is 4.5, with a rising trend over the first three years. Panel (b) of Figure 3 reports the impact of HDS entry on formal retail employment. The coefficients range from an increase of less than 10% to one of approximately 40% after the six years following the first opening. There is evidence of a slight pre-treatment trend in formal retail employment using administrative data, but as formal retail represents less than one percent of baseline formal employment, it does not materially affect the subsequent post-treatment results (the observed pre-trend also flattens when we restrict our estimation sample to the municipalities covered in the survey data). This indicates that the direct demand for formal retail jobs from HDS outweighs any potential job losses in incumbent retail firms, such as supermarkets, due to increased competition within the formal, modern retail sector.[21] Together, these results can be interpreted as first-stage effects: entry triggers subsequent store expansion and generates positive changes in formal retail employment relative to pre-entry levels.

[21]Ideally, we would want to separate direct and indirect effects, the former being how many jobs are generated directly by HDS. However, when reporting payroll on the PILA, discount chains often assign employees to the nearest capital city (i.e., instead of reporting payroll in the municipality where the store is located, they report it in the closest large urban area). Overall, fewer than 20% of municipalities with hard discount chains report formal employees in PILA.

Figure 3: Direct Effects of HDS



(a) Number of stores

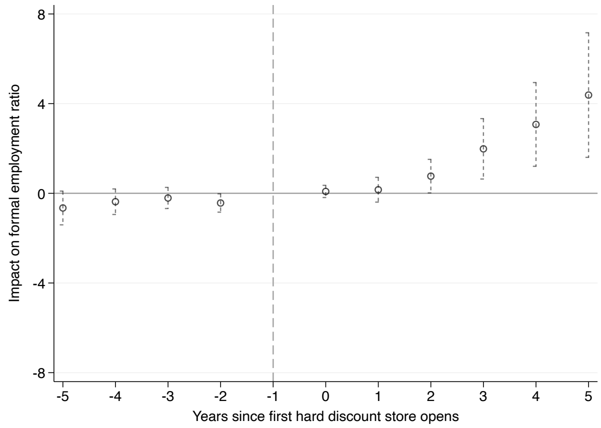
(b) Formal retail employment

Note: We use the *C&S* estimator. The dependent variable in panel (a) is the number of stores, and in panel (b) is retail formal employment over the working-age population as defined in the 2005 Census. Both regressions are weighted by the local working-age population in 2005. Panel (b) coefficients are expressed in percentage changes, obtained by normalizing estimates by the pre-treatment mean of the outcome. The number of treated municipalities is 372 in PILA and 191 in GEIH. Standard errors are clustered at the municipality level. 90% confidence intervals. Source: PILA 2010–2018 (August) and GEIH 2010–2018.

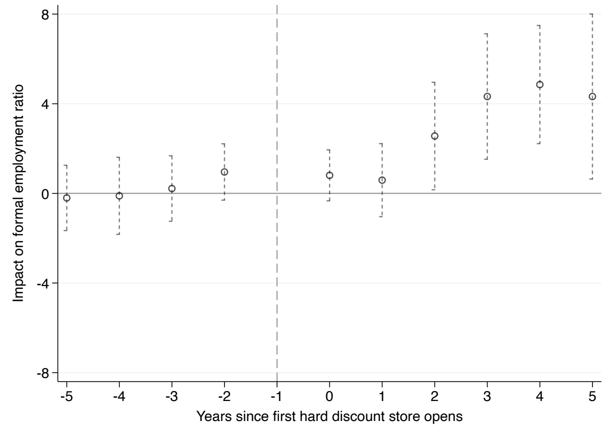
5.1 Employment effects

We now analyze the evolution of formal employment in local labor markets following store openings. The main finding of the paper appears in Figure 4. It shows that formal employment increases substantially after the arrival of discount stores, according to both administrative and survey data, and that this effect takes three years to materialize, in line with the number of stores rising in the municipality. Averaging all six post-treatment periods, the ratio of formal employment-to-population increases by around 1.7 pp using administrative records and 2.9 pp using survey data. Importantly, treated and not-yet-treated municipalities do not exhibit divergent trends prior to treatment, suggesting that the arrival of HDS is unrelated to local employment trends across these groups.

Figure 4: Event study estimates on formal employment



(a) Formal employment (Admin)



(b) Formal employment (Survey)

Note: We use the *C&S* estimator. The dependent variable in panel (a) is formal employment using PILA, and in panel (b) is formal employment using GEIH, both divided by the working-age population according to the 2005 census. Regressions are weighted by the local working-age population in 2005. The number of treated municipalities is 372 in PILA and 191 in GEIH. Standard errors are clustered at the municipality level. 90% confidence intervals. The coefficients represent percentage point changes. Source: PILA 2010-2018 in August and GEIH 2010-2018.

To benchmark the estimates, the weighted mean of the formal employment-to-population ratio before the treatment is 16% in the administrative data and 28.3% in the survey. Hence, the increases in relative terms are 10.6% and 10.3%, respectively, which are fairly similar across datasets. These results remain robust when controlling for baseline covariates (such as distance to major capital cities and a proxy for economic development) and when accounting for violations of parallel pre-trends, and remain statistically significant under randomization of treatment timing. Appendix Table E.1 shows that the estimates are also similar when we control for baseline informality.

Since the sample of treated and control municipalities is unbalanced in event time, we further decompose these coefficients by treatment cohort. To that end, we use the properties of the *C&S* estimator, which can flexibly aggregate effects into different components and aggregate by treatment cohort rather than by event time. Appendix Table F.5 shows that the effect is concentrated in the municipalities that opened before 2016. This is expected, as these municipalities have been observed for at least 3 years, during which more stores have opened. This distinction is important for extrapolating effects, indicating that not all future-treated municipalities will experience outcomes identical to those of the initially treated cohort. Altogether, discount stores help formalize local economies and generate employment growth, partly by increasing the employment rate and reducing inactivity and unemployment.[22]

Because these coefficients measure the overall impact of the first opening, we also use the number of stores in each municipality to quantify the intensive-margin effect. From the earlier 4.5 average number of stores in our estimation window, we calculate a per-store impact: each additional store increases the formal employment ratio by 0.4 pp (= 1.7/4.5), or 2.5%, using administrative data, and by 0.6 pp (= 2.9/4.5), or 2.1%, using survey data relative to the baseline mean. When the number of jobs is used as the outcome, each store generates approximately 110 direct and indirect formal jobs, implying a job multiplier of about 11 (=110/10).

Discussion of magnitudes. The formal employment multipliers we estimate are higher than those for the US, where one additional manufacturing job generates approximately 1.6 jobs in the non-tradable sector (Moretti, 2010). This is consistent with our sample consisting of relatively small geographic units, averaging fewer than 7,000 formal workers (Appendix Table F.1), which likely have a more elastic local labor supply than U.S. metropolitan statistical areas. In line with this, estimated impacts are more pronounced in less developed municipalities without survey coverage (Appendix Figure F.4), supporting the view of a more elastic local labor supply in smaller, less populated, and more informal areas.

The impact on formal employment also reflects the scale of the economic shock they represent in the small urban municipalities we study. As discussed in Section 2, investment per store ranges from about 154,000 USD to nearly 1 million USD, depending on the discount chain, equivalent to roughly 4 to 19 percent of average annual municipal government investment. Furthermore, to benchmark the scale of the retail demand shock associated with store entry, we compare annual

[22] Appendix Figure F.3 illustrates the effects on the local employment rate. We find positive estimates that grow over time. Over the six post-treatment years, HDS openings increased the local employment rate by approximately 2.3 pp. Additionally, Appendix Table F.4 shows a significant reduction in the inactivity and unemployment rate following store openings.

gross sales of hard discount chains with municipal value added. Their gross sales amount to around 5 percent of average municipal value added.[23] This benchmark captures the scale of stores' own sales activity and associated demand for products sold at HDS, not their direct contribution to municipal GDP through their value added. Given the high levels of informality in our sample, the relative importance of this shock within the formal economy is likely even larger, making a 10 percent increase in formal employment economically plausible.

Lastly, this employment effect may be generated not only by demand within the treated municipality. Once a hard discount chain contracts a local supplier, its procurement model typically involves buying in bulk to supply stores across multiple cities, so part of the estimated effect may instead reflect national, rather than only local demand, which could also explain why our implied formal employment multiplier is relatively larger.[24]

Heterogeneity by industry sectors. Panels (a) and (b) of Figure 5 indicate that the primary and secondary sectors account for most of this growth, according to both administrative and survey data. Focusing on survey data, which provides a more accurate industry classification, the manufacturing employment-to-population ratio rises by 0.9 pp, the agricultural employment ratio by 0.6 pp, and the construction ratio by 0.4 pp, explaining around 65% of the overall increase (Appendix Tables F.6, F.7, F.8, and F.9 provide detailed point-estimates).

There are several plausible explanations for these inter-sectoral spillovers from retail, operating through both firm-level and consumer-level channels. On the firm side, hard discount chains have strong incentives to build local supplier networks for intermediate inputs, since transportation costs in Colombia are among the highest in the world (Londoño-Kent, 2009). This reliance on

[23] We compute this ratio by dividing estimated municipal hard discount sales by municipal value added. According to Euromonitor International (2023a), average annual sales per store between 2017 and 2022 were 1.38 million USD, which, multiplied by an average of 4.5 stores, yields 6.21 million USD in annual sales per municipality. Data from DANE indicate that the average municipal value added over the same period was approximately USD 123 million.

[24] We have suggestive evidence of regional demand still not being the main driver of our results: excluding municipalities in commuting zones around capital cities (Appendix Table E.6) and controlling for driving time to Bogotá and Medellín (Appendix Figures E.2 and E.3) leave our estimates around similar magnitudes. This indicates that demand does not meaningfully change within commuting zones or when controlling for proximity to the largest cities, although we cannot rule out the possibility that local suppliers serve national demand in ways not captured by these network proxies.

nearby producers can raise demand for formal labor in upstream manufacturing and related industries.[25] Unfortunately, we cannot directly measure these linkages because we lack input-output data. Anecdotally, the largest hard discount chain reported that around 80% of their goods in 2020 were produced in Colombia (Asmar Soto, 2020), and according to a discount chain manager, many suppliers have expanded capacity to meet the increased demand generated by these stores (Acosta, 2018). Firm-level spillovers may also arise through local investment. The rapid expansion of hard discount chains requires substantial construction activity, which can generate employment effects in construction and related sectors. In addition, their presence may create agglomeration forces that benefit nearby formal businesses (Evensen et al., 2023).

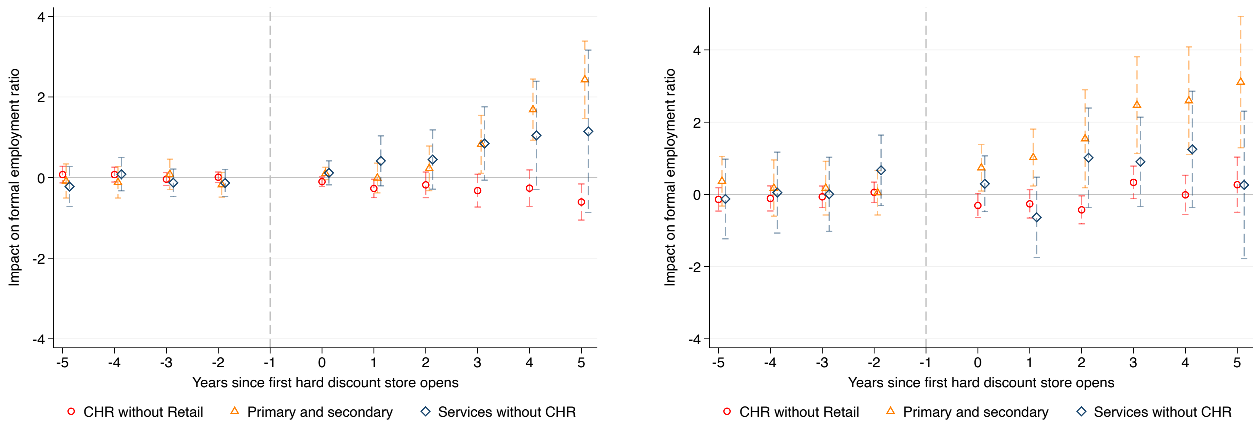
On the consumer side, HDS can affect local economies by lowering grocery prices and increasing real incomes. Lower food expenditures and higher earnings for new formal labor market entrants employed by these chains may stimulate local demand for goods and services, generating demand-driven multiplier effects that operate primarily through household consumption (Gerard et al., 2026).

Identifying the dominant channel is challenging, as these mechanisms are likely interlinked. However, the sectoral evidence points more strongly toward firm-level channels, particularly upstream supply chain effects and local investment. Two findings support this interpretation. First, if consumer-driven multipliers or agglomeration effects were the primary mechanism, we would expect more diffuse gains across sectors, including the informal sector, which represents a large share of employment in these municipalities. As shown later, we do not observe positive effects in informal employment or wages. Second, although HDS provide non-tradable retail services, they sell tradable goods and operate as distribution platforms tightly connected to domestic supply chains. Given the larger effects in agriculture and manufacturing, this suggests that supply chain linkages seem to play a more central role. We also find positive effects in construction, consistent with the investment channel driven by rapid store expansion.

[25]These findings are consistent with de Paula and Scheinkman (2010), who discuss the ”business formality chain” that emerges when increased enforcement at the production stage, combined with value-added tax collection, incentivizes formality both upstream and downstream in the supply chain.

Additional support comes from Atkin et al. (2018), who find no local employment effects from the entry of foreign retailers in Mexico, despite their much larger scale. These retailers import most of their products from the US or Europe, limiting spillovers to domestic firms. In contrast, the positive employment effects we document are consistent with discount chains' heavier reliance on local suppliers, which strengthens firm-level spillovers within the domestic economy.

Figure 5: Event study estimates on formal employment ratios by industry



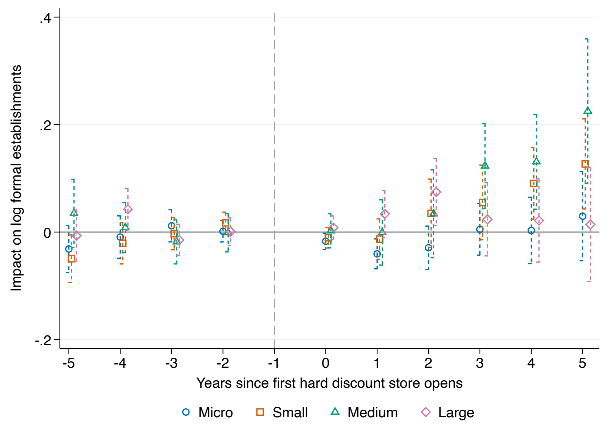
(a) Formal employment (Admin)

(b) Formal employment (Survey)

Note: We use the *C&S* estimator. The dependent variable in panel (a) is formal employment in each sector using PILA, and in panel (b) is formal employment in each sector using GEIH, both divided by the working-age population according to the 2005 census. CHR refers to commerce, hotels, and restaurants. The primary and secondary sectors comprise manufacturing, construction, agriculture, and mining. Regressions are weighted by the local working-age population in 2005. The number of treated municipalities is 372 in PILA and 191 in GEIH. Standard errors are clustered at the municipality level. 90% confidence intervals. The coefficients represent percentage point changes. Source: PILA 2010-2018 in August and GEIH 2010-2018.

Number of formal establishments. Consistent with the firm-side channel, we find that HDS entry increases the number of formal small- and medium-sized establishments by 4.7 and 8.6 percent, respectively (10–50 employees for small and 51–200 for medium-sized), as reported in Appendix Table B.1, based on social security records. Importantly, Figure 6 shows no evidence of pre-treatment entry into formality and no increase during the first three years after HDS entry. The rise in establishments occurs only thereafter, in line with the expansion in the number of stores. We also find no effects for micro (2–9 employees) or large (200+ employees). These patterns are consistent with HDS entry fostering the expansion of some local establishments, with growth in formal employment reflecting both incumbent firm adjustments and entry into formality.

Figure 6: Event study estimates on log formal establishments by firm size



Note: We use the *C&S* estimator. The dependent variable is the log number of formal establishments by size category. Firm size categories are defined as micro (2–9 workers), small (10–50), medium (51–200), and large (200+). Regressions were weighted with the local working-age population in the 2005 census. Standard errors are clustered at the municipality level. 90% confidence intervals. The coefficients represent log points. Source: PILA 2010–2018.

Informal employment. The main competitors of HDS are local neighborhood shops that are primarily informal. To analyze this outcome, we can rely only on the GEIH survey data. Appendix Table E.3 shows that the impact on the informal employment-to-population ratio is not robust, as the coefficient ranges from negative to positive values when we include baseline covariates. Therefore, the impact on informal employment is imprecise. Regarding sectoral impacts, increased competition between HDS and the informal retail sector does not affect employment growth in either sector. In fact, the point estimate is positive across different specifications (see Appendix Table E.3). Lastly, the coefficients on informal employment across other sectors vary substantially but are too noisy to describe a clear pattern (see Appendix Table F.10).

A limitation of using the GEIH survey is that it does not cover all the municipalities where HDS operate. To address this, we leverage Colombia's social security system to obtain complementary measures of the population outside of the formal sector. In Colombia, universal access to the health system is achieved through subsidies or contributions. Formal workers contribute to the system, while subsidized beneficiaries are either informally employed, unemployed, or not in the labor force. Using the beneficiaries of subsidized social protection as a proxy for the population outside the formal sector, we show in Appendix Figure F.5 that in the first four years of the treatment, the effect is close to zero. However, in later years, we observe a negative trend consistent with the observed increase in formality. Overall, the main impact of HDS is a substantial shift in the workforce composition, with formality driving the increase in employment.

5.2 Taxes

The arrival of HDS boosts local economic activity through direct job creation, spillover effects on other sectors, and agglomeration effects in areas where the stores are located. We use municipality-level tax revenues to explore whether HDS entry has broader impacts on local tax revenues. Although the central taxes in Colombia, such as income tax, social security contributions, and VAT, are collected nationally, other significant taxes are collected at the municipal level. These include property, industry, commerce, and vehicle and gasoline taxes.[26] Industry and commerce taxes, in particular, are highly correlated with the performance of industrial, commercial, or service activities registered in the municipality. Therefore, even though HDS chains operate nationwide, they are required to pay industry and commerce taxes based on local sales and revenues in each municipality where they have stores.

Figure 7 shows that before the arrival of HDS, the growth of taxes-to-baseline public revenues was similar between the early and late treated municipalities.[27] After the entry of discount stores, the ratio of local tax revenues increases by 10.1 pp on average, or 7.5% relative to the pre-treatment mean, for all post-treatment periods.[28] Most of this increase is attributed to revenues collected from industry and commerce taxes (Appendix Table F.11 provides detailed point estimates). Hence, the arrival of HDS substantially boosts tax revenues for local governments, primarily driven by the

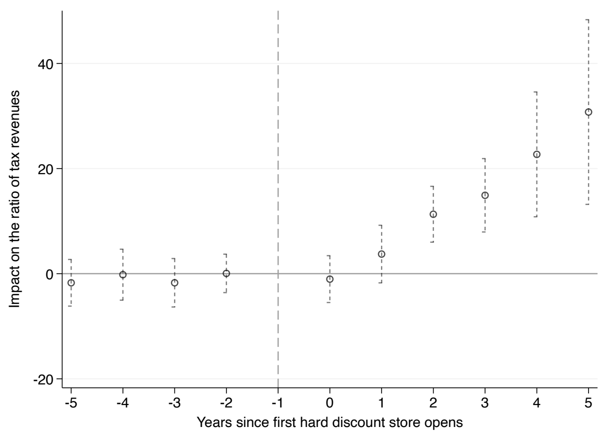
[26] The owners of low-income residential households are not required to pay property tax.

[27] A potential concern is that local policymakers might reduce municipal taxes to attract discounters and other investors. However, tax rates remained largely stable during our study period (2010–2018).

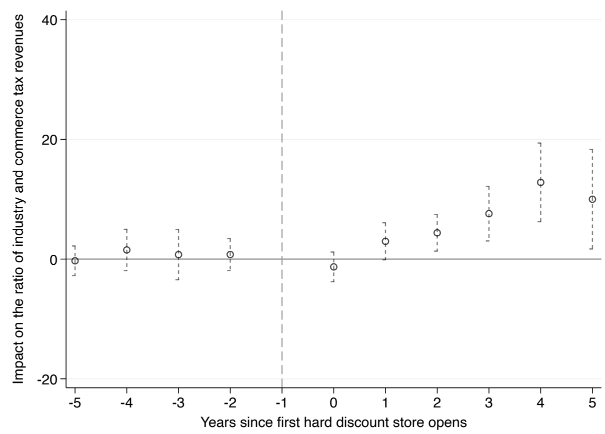
[28] We use pre-shock local tax revenues as regression weights, rather than population, to capture differences in the size and capacity of municipal tax authorities.

growth in manufacturing and commercial activities. Increasing local revenues could trigger local public spending, which, in turn, could stimulate the labor market, indicating another mechanism through which HDS could promote local formal employment. These results align with previous findings by sector and with the literature on the supply chain effects of business formality (de Paula and Scheinkman, 2010; Gerard et al., 2023; Rios and Setharam, 2018), indicating that HDS suppliers are likely to be formal if they seek to claim all tax benefits and discounts available under the law.

Figure 7: Event study estimates on the ratio of taxes by type



(a) Revenues of all taxes



(b) Revenues of industry and commerce taxes

Note: We use the *C&S* estimator. The dependent variable is revenue from each specific type of local tax divided by total revenues, including taxes and central government transfers. We include only taxes collected at the municipal level, such as property, industry and commerce, gasoline, and vehicle taxes, as well as other local taxes, such as the right to post advertisements on public streets. Regressions are weighted by municipalities' 2010 revenues. The number of treated municipalities is 371. Standard errors are clustered at the municipality level. 90% confidence intervals. The coefficients represent percentage point changes. Source: DNP, 2010-2018.

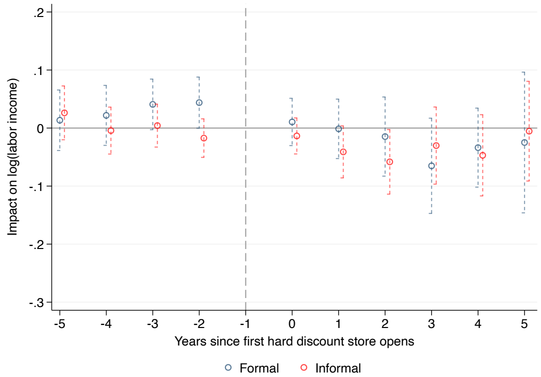
5.3 Labor Income and Working Hours

Other employer margins of adjustment, such as on labor income or working hours, may occur in response to the hard discount store expansion. One limitation of our measure of these outcomes is that it captures only local averages across all workers, as compositional changes over time are not accounted for. We first focus on the impacts on labor income across the formal and informal sectors. Figure 8a shows a slight negative effect in the first post-treatment years on the labor income of informal workers, while the impact on the labor income of formal workers is rather stable. Moreover, using administrative data, we find that formal wages have an effect close to zero that turns negative in the last post-treatment period (see Appendix Figure F.6). Hard discount firms pay higher wages, which should increase average formal wages, but because the primary and secondary sectors drive the main increase in formal employment, they likely hire minimum-wage workers, thereby decreasing average formal wages. Another potential explanation is that local labor supply is highly elastic, which mutes any formal wage effect (Moretti, 2010).

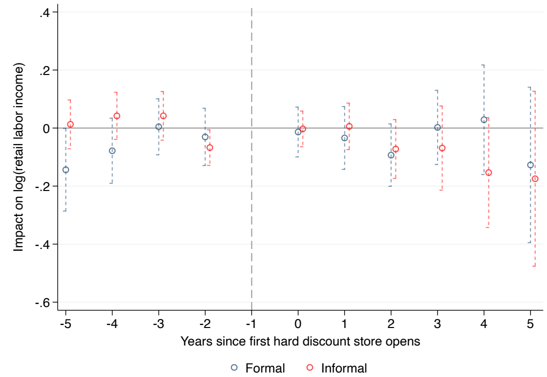
We now measure the impact on retail workers' labor income by formality status. For informal workers, this serves as an indirect measure of neighborhood shop owners' earnings, while for formal workers, it reflects regular wages. Figure 8b shows an imprecise negative growth in the labor income of informal workers in the latest post-treatment periods, indicating that the increase in competition from HDS may affect them mostly through reduced earnings rather than employment losses. Due to the nature of their labor contracts, informal shop owners do not pay the mandatory minimum wage to their workers, thereby having the flexibility to adjust to demand shocks. While the estimates are insignificant and imprecise, they are notably large ($ATT_{post} = -7.8\%$). Regarding formal retail workers, even though these firms pay higher premiums nationwide, this does not translate into higher average local wages.

Lastly, we focus on average weekly working hours across the informal and formal sectors, as this is another possible margin of adjustment for employers. For instance, discount stores may have longer opening hours to attract more customers, so other businesses might adjust their hours. Figure 8c shows that there are no significant changes in working hours after the first entry.

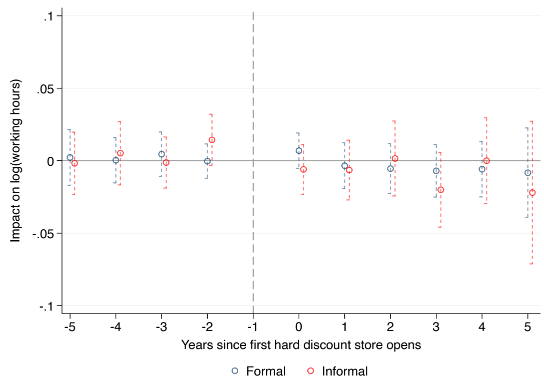
Figure 8: Event study estimates on labor income and working hours by sector



(a) Labor income



(b) Retail labor income



(c) Working hours

Note: We use the *C&S* estimator. The dependent variables are the logarithms of formal and informal labor income and working hours. Regressions are weighted by the local working-age population in the 2005 census. The number of treated municipalities is 191. Standard errors are clustered at the municipality level. 90% confidence intervals. The coefficients represent log points. Source: GEIH 2010-2018.

6 Robustness Checks

We assess the robustness of our main results along three dimensions. First, we examine sensitivity to additional baseline covariates and potential confounding factors, including migration and proximity to capital cities. Second, we evaluate alternative specifications, varying the regression weights, the functional form of outcomes, and the treatment of outliers. Third, we test the identifying assumptions using placebo exercises and sensitivity analyses for deviations from parallel trends.

Our results are robust to the inclusion of baseline covariates and controls for potential confounders, including migration and proximity to capital cities. They also hold under alternative specifications that vary regression weights, the functional form of outcomes, and the treatment of outliers, and remain statistically significant under placebo exercises that randomize treatment timing and under sensitivity analyses that allow for moderate violations of parallel pre-trends, though not for larger ones. See Appendix E for the complete discussion and results.

7 Conclusion

In this paper, we measure the impact of the expansion of Colombia's leading hard discount chains on local labor markets. Unlike previous studies focusing on large retailers such as Walmart or Amazon FCs, we examine the effect in a developing country with a large informal sector. Our findings reveal notable shifts in local labor markets following the introduction of HDS. Notably, HDS entry increases local formal employment by around 10%, driven by manufacturing, construction, and retail. This suggests that the entry of hard discount chains into a municipality generates strong spillover effects from retail to other sectors. A potential mechanism is the increased demand for some of the goods that local formal firms produce within the discounters' supply chain, prompting supplier firms to hire more formal workers to meet the higher demand. Importantly, the employment effects are not immediate, appearing up to 3 years after the first opening of HDS, aligning with the time and investments required for a hard discount chain to attract more customers and expand within the treated municipalities.

Regarding the informal retail sector, the findings suggest that discount stores do not affect employment in this sector. However, we cannot rule out whether HDS entry leads to the closure of local neighborhood shops due to a lack of more precise data. At the same time, labor earnings in the informal retail sector exhibit a noisy negative trend after the entry of hard discounters. We rationalize that the adjustment may occur through earnings rather than employment, as neighborhood shops are typically small, entrepreneurial operations run mostly by their owners, with only 16% employing additional workers.

Lastly, we find suggestive evidence that the positive employment impacts contribute to aggregate tax effects in local economies. After the entry of discounters, the ratio of collected taxes to total public revenues increases by an average of 7.5%, driven by revenues from industry and commerce taxes. Collectively, these findings have important policy implications for developing countries, such as promoting a specific type of business that may spur the formalization of local economies alongside other public policies. Further research is needed to better understand the potential effects on neighborhood shop profits and the longer-term implications of the expansion of hard discount chains in developing countries.

Acknowledgments: We are thankful to Luis Eduardo Arango, Jhorland Ayala, Jaime Bonet, Leonardo Bonilla, Juan Esteban Carranza, Jorge Flórez, Luis Armando Galvis, Sergio Ocampo, Miguel Talamas, Jan Stuhler and Gabriel Ulyssea, as well as participants at the NCDE in Oslo, 49th Symposium of the Spanish Economic Association, CEDE Seminar at Universidad de los Andes, UC3M Applied Reading Group, LACEA-LAMES 2023, Red Investigadores de Economía and Banco de la República seminar for helpful comments and discussions. We thank María Camila Gómez and Julián Gallego Villa for providing excellent research assistance. The views and opinions expressed here are those of the authors and do not necessarily reflect the views of Banco de la República or its Board of Directors. Any errors are exclusively the responsibility of the authors. This paper was previously circulated as: "The Impact of Hard Discount Stores on Local Labor Markets: Evidence from Colombia".

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Online Appendix

A Neighborhood shops characteristics

We analyze information on neighborhood shops by exploiting a micro-business survey that contains informal and formal firms (EMICRON, by its acronym in Spanish).[A.1] These small local grocers represent a crucial channel for the grocery retail industry and a relevant source of informal employment. Table A.1 shows that 88% of the surveyed businesses are run by their owners, who are also the sole employee, and that 44% of workers in businesses with more than one employee are unpaid, usually relatives.[A.2] These businesses also have lower survival rates than other micro-businesses in the survey: almost one-third have been operating for at least 10 years, compared with 48% of businesses in other sectors.

Neighborhood shops are highly informal. Only 17% of them have an up-to-date registration with the local chamber of commerce, and fewer than 6% file any tax return.[A.3] Regarding employment informality, almost 9 out of 10 business owners do not contribute to the social security system. For businesses with employees, the informality rate is also around 91%. This is considerably higher than Colombia's overall employment informality rate in 2019, which was 50%.

The average neighborhood store sells about US$ 717 per month in merchandise and sells about US$1,000 per month. Other monthly expenses (such as utilities, rent, and transportation) total US$100. For stores with paid employees, the average monthly wage is approximately US$200, which is 30% below the mandatory monthly minimum wage for 2019. This results in an average monthly profit of around US$380. When self-reporting their average monthly profits, the typical response is around US$215.

There are important differences in the descriptive analysis by formality status. Table A.1 reports the mean and standard error of key indicators for two subgroups: stores with an updated register at the local chamber of commerce (which we classify as formal) and stores without one (which we classify as infor-

[A.1]The survey was conducted in 2019, and it only includes businesses with up to 10 employees. We have data at the two-digit industry level, so we identify neighborhood shops as businesses under ISIC Revision 4 code 45: "Wholesale and retail commerce and repair of motor vehicles and motorcycles". Throughout the analysis, we use an exchange rate of 3,250 Colombian pesos per US dollar to convert the financial variables. This is an approximation of the average USD/COP rate in 2019.

[A.2]In all results, we use the survey's sampling weights to compute the reported means and standard deviations.

[A.3]Three taxes are particularly important for businesses in Colombia: the VAT, the income tax, and the industry and commerce tax. The first two are at the national level, whereas the last is paid to the municipality.

mal). We plot some of these results in Figure A.1. Panel (a) shows that, on average, formal stores have merchandise sales 3.5 times higher than those of informal stores. They also have higher merchandise costs: 6.5 million pesos for the typical formal store (US$2,000), compared with 1.5 million pesos for the typical informal store (US$450). Thus, formal stores are, on average, more profitable than informal stores: profits per worker from merchandise sold are US$187 per month in the typical informal store and US$553 per month in the typical formal store.

Panel (b) of Figure A.1 shows differences in business characteristics between formal and informal stores. Formal shops have larger staffs than informal ones and rely less on unpaid workers. They are less likely to have started out of necessity and more likely to have started in response to a business opportunity. They are also more likely to survive in the market.[A.4] Regarding employment informality, almost 4 in 10 formal business owners contribute to the social security system, compared with 1 in 10 for informal businesses, and the average share of informal paid workers is 80% for formal stores, versus 97% for informal stores.

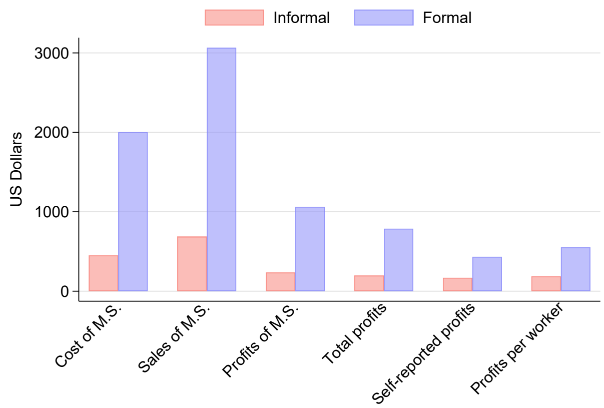
[A.4] Stores with an updated register at the Chamber of Commerce are also more likely to be classified as formal using other definitions: about 30% of them report income, VAT, or commerce tax, compared with only 1.2% of informal stores.

Table A.1: Summary statistics of corner shops by formality status

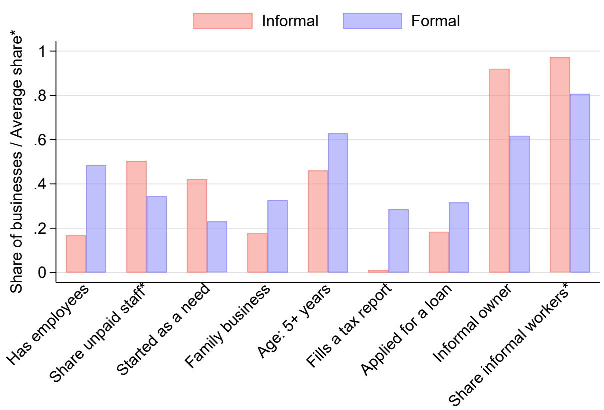
	Total sample mean (S.D.)	Formal mean	Informal mean
Panel A: Basic business characteristics			
Has employees	0.222	0.485	0.168
	(0.415)	(0.500)	(0.373)
Total business staff	1.320	1.832	1.215
	(0.744)	(1.210)	(0.549)
Share of unpaid staff in personnel	0.444	0.344	0.504
	(0.486)	(0.456)	(0.493)
Owner started bussiness alone	0.736	0.595	0.765
	(0.441)	(0.491)	(0.424)
Family business	0.204	0.326	0.179
	(0.403)	(0.469)	(0.384)
Business started as a need	0.389	0.230	0.422
	(0.488)	(0.421)	(0.494)
Business age: less than a year	0.164	0.070	0.183
	(0.370)	(0.255)	(0.387)
Business age: more than 1 year and less than 5 years	0.347	0.302	0.356
	(0.476)	(0.459)	(0.479)
Business age: more than 5 years	0.489	0.628	0.461
	(0.500)	(0.483)	(0.498)
Business located in household dwelling	0.332	0.306	0.338
	(0.471)	(0.461)	(0.473)
Panel B: Informality of the business and the owner			
Business reports income, VAT, or commerce tax	0.059	0.286	0.012
	(0.235)	(0.452)	(0.109)
Business applied for a loan	0.207	0.316	0.184
	(0.405)	(0.465)	(0.388)
Owner does not contribute to health or pension	0.869	0.618	0.921
	(0.337)	(0.486)	(0.270)
Panel C: Costs, sales, and profits (in USD)			
Cost of merchandise sold during last month	717.293	2,001.617	451.123
	(2,212.441)	(4,379.250)	(1,233.752)
Merchandise sales during last month	1,095.262	3,065.198	687.003
	(2,918.777)	(5,787.930)	(1,539.870)
Profits for merchandise sold during last month	377.970	1,063.581	235.880
	(1,090.237)	(2,046.299)	(670.479)
Total profits during last month	297.626	787.681	197.162
	(963.807)	(1,799.183)	(629.619)
Self-reported average monthly profits	214.648	434.129	169.653
	(486.733)	(745.249)	(399.647)
Average merchandise profits by employee	250.576	553.774	187.739
	(602.826)	(924.643)	(488.391)
Panel D: Personnel characteristics			
Share of women in personnel	0.541	0.560	0.529
	(0.466)	(0.446)	(0.477)
Average employee tenure in months	56.791	61.062	54.256
	(78.647)	(80.189)	(77.606)
Share of fully-informal employees	0.912	0.808	0.973
	(0.272)	(0.376)	(0.154)
Average wage (for paid employees) in USD	199.282	227.973	164.504
	(104.991)	(88.772)	(112.363)
Maximum obs (unweighted)	22,675	3,994	18,681
Maximum weighted obs	1,408,925	239,698	1,169,227

Note: This table shows the mean of corner shops by formality status. A business is formal if it maintains an up-to-date registration with a chamber of commerce. We use a USD-to-COP exchange rate of 3,250 (which approximates the average rate in 2019) and define corner shops as businesses classified under ISIC Revision 4 code 45: "Wholesale and retail commerce and repair of motor vehicles and motorcycles". The mean and standard deviation are weighted using the survey's sample weights. Standard deviations are in parentheses. Source: 2019 Micro Businesses Survey (EMICRON), DANE.

Figure A.1: Neighborhood shop characteristics by formality status



(a) Average monthly costs of merchandise sold (M.S), sales, and profits



(b) Employment, business origin, and other definitions of informality

Note: This figure reports the mean of selected characteristics for corner shops by formality status, using the 2019 Micro Businesses Survey. A business is formal if it maintains an up-to-date registration with a chamber of commerce. We use a USD-to-COP exchange rate of 3,250 (which approximates the average rate in 2019) and define corner shops as businesses classified under ISIC Revision 4 code 45: "Wholesale and retail commerce and repair of motor vehicles and motorcycles". The mean and standard deviation are weighted using the survey's sample weights.

B Discussion of identification assumptions

A potential threat to our identification is a time-varying shock to outcomes that is correlated with the *timing* of discounters' entry, which could confound our estimates. For instance, the entry of discount stores might coincide with the entry of other formal firms in the municipality. Figure 6 shows no pre-treatment differential establishment entry and no increase in the first years after HDS entry, with positive effects emerging only after the third year for smaller and medium-sized establishments, consistent with store expansion (see Appendix Table B.1 for average pre- and post-treatment estimates). A related concern is that hard discount chains tend to open first in places closer to and better connected to capital cities. Consequently, early-treated cohorts could be more affected by spillover trends from capital cities than later-treated cohorts. To address this concern, we adjust our empirical strategy to include baseline controls, thereby assuming conditional parallel trends, under which treated and not-yet-treated municipalities with similar covariate values would have evolved in parallel in the absence of the treatment. With this weaker identifying assumption, we are comparing treatment cohorts of municipalities with similar baseline distances to the largest cities of the country, Medellín and Bogotá, and with similar development stages (measured with the rurality index) to show that results are robust and do not change significantly, albeit specific outcomes become more imprecise (this is expected as the size of certain cohorts is not large). For this, we use the outcome regression adjustment in the *C&S* framework. Lastly, we assume there are no anticipatory effects in response to store openings. Given that the shock is relatively unexpected and its effects take years to materialize, this is less of a concern.

Another relevant aspect is the treatment's intensive margin across municipalities. If cohorts are heterogeneous in the number of stores they have, interpreting the overall ATT becomes less straightforward. Appendix Figure C.2a presents evidence that cohorts receive a similar number of stores over time. In all cohorts, half of the municipalities have two discount stores three years after the arrival of discount chains. The top 20% of municipalities, by the number of stores received, have four, while those that receive the fewest have one. When differentiating by cohort, the average number remains similar across cohorts. The only notable exception is the 2016 cohort, which saw the arrival of *Justo & Bueno*. Even in this case, the difference is about one store more on average (see Appendix Figure C.2b).

We also discuss the choice to use the not-yet-treated municipalities as our main control group rather than the never-treated (that is, municipalities that had not received a hard discount chain as of 2020). We argue that the never-treated group is not a suitable control group given the substantial differences between these municipalities and those where hard discount chains decided to establish stores. As shown in Appendix Table C.1, municipalities that received at least one HDS were more populous, less rural, and closer to the main municipalities of Bogotá and Medellín. Furthermore, we constructed a propensity score using a logit regression to predict the likelihood of treatment based on pre-treatment characteristics, and found that the distributions are highly skewed, highlighting the stark differences between the two sets of municipalities (see Appendix Figure B.1).

Apart from cross-sectional differences in pre-treatment characteristics, the labor markets of never-treated municipalities exhibit trends that differ from those of treated areas. Panel B of Appendix Table B.2 shows that when we use the never-treated group as a control, significant pre-treatment trends emerge in employment, unemployment, and inactivity rates. In contrast, comparing treated and not-yet-treated municipalities yields pre-treatment coefficients that are statistically indistinguishable from zero for all these outcomes. Moreover, in Panel C, we restrict the sample of never-treated municipalities to those similar to treated municipalities by employing a one-to-one propensity score matching model without replacement. Even in the restricted never-treated group, the pre-trends persist, albeit smaller in magnitude and not statistically significant. Together, these findings indicate that never-treated municipalities differ from treated municipalities in unobserved trends and thus are not an appropriate counterfactual for our analysis.

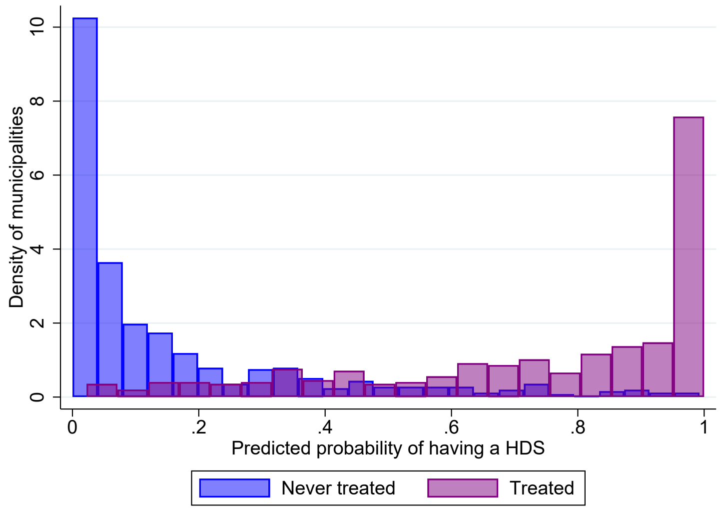
A similar problem arises when we include department capitals in our estimation sample. Panel B of Appendix Table B.3 shows the C&S estimates for formal employment by sector using the administrative data. There are significant pre-trends in total formal employment and in the retail, primary, and secondary sectors. This does not occur when we exclude capital cities (as in Panel A). Therefore, we report results excluding capital cities.

Table B.1: Average *C&S* estimates of log formal establishments by size

	(1) Micro	(2) Small	(3) Medium	(4) Large
ATT_{pre}	-0.007	-0.014	0.006	0.005
	(0.018)	(0.017)	(0.024)	(0.018)
ATT_{post}	-0.008	0.047*	0.086**	0.029
	(0.024)	(0.028)	(0.038)	(0.030)
N	3,348	3,347	3,330	3,346
Municipalities	372	372	370	372

Note: The dependent variable is the log number of formal establishments by size category. Size categories are defined as micro (2–9 workers), small (10–50 workers), medium (51–200 workers), and large (200+ workers). Regressions were weighted by the local working-age population in the 2005 census. Standard errors are clustered at the municipality level. Coefficients are in log points. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: PILA 2010–2018.

Figure B.1: Distribution of propensity scores by treatment status



Note: This figure plots the distribution of the predicted probabilities of receiving HDS before 2020 by treatment status. We estimate a logit model with independent variables matching those in Table C.1 (excluding department dummies). We then predict the probability at the municipality level, which is depicted on the x-axis.

Table B.2: Average *C&S* estimates of labor market rates using never-treated municipalities as control group

	(1) Employment rate	(2) Unemployment rate	(3) Inactivity rate
Panel A: Not-yet-treated as control group			
ATT_{pre}	0.718	-0.477	-0.439
	(0.536)	(0.398)	(0.443)
ATT_{post}	2.286***	-0.990	-1.886**
	(0.849)	(0.628)	(0.728)
$\text{ATT}_{post_{k=3,4,5}}$	3.242***	-1.758**	-2.369**
	(1.191)	(0.884)	(1.058)
Panel B: Never treated as control group			
ATT_{pre}	1.737***	-1.201***	-1.022**
	(0.521)	(0.388)	(0.435)
ATT_{post}	-0.234	0.404	0.003
	(0.534)	(0.470)	(0.492)
$\text{ATT}_{post_{k=3,4,5}}$	-0.036	0.409	-0.152
	(0.711)	(0.649)	(0.663)
Panel C: Never treated as control group (restricted)			
ATT_{pre}	1.153	-0.204	-1.163
	(0.919)	(0.751)	(0.730)
ATT_{post}	1.367	0.873	-2.163*
	(0.996)	(0.785)	(1.152)
$\text{ATT}_{post_{k=3,4,5}}$	2.824*	0.860	-3.856**
	(1.645)	(1.095)	(1.830)

Note: The dependent variable in each column is the yearly labor market rate. The sample of never-treated municipalities in Panel C is restricted to those selected via one-to-one logit-based propensity score matching using the dependent variable and the covariates in Column 7 of Table C.1. Regressions were weighted by local employment in 2010, and standard errors are clustered at the municipality level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: GEIH 2010-2018.

Table B.3: Average *C&S* estimates of formal employment ratios by sector including capital cities

	(1) Total	(2) Retail	(3) CHR without retail	(4) Primary and secondary	(5) Services without CHR
Panel A: Excluding capital cities (Baseline)					
ATT_{pre}	-0.417	-0.045*	0.030	-0.074	-0.100
	(0.294)	(0.025)	(0.090)	(0.205)	(0.211)
ATT_{post}	1.742***	0.108**	-0.290*	0.877***	0.670
	(0.608)	(0.044)	(0.158)	(0.296)	(0.474)
$\text{ATT}_{post_{k=3,4,5}}$	3.147***	0.170**	-0.397*	1.647***	1.013
	(1.049)	(0.066)	(0.228)	(0.426)	(0.756)
Panel B: Including capital cities					
ATT_{pre}	-1.676***	-0.104***	0.212*	-0.585**	-0.705
	(0.607)	(0.039)	(0.112)	(0.292)	(0.454)
ATT_{post}	6.523***	0.292***	-0.775***	1.639***	4.108**
	(2.358)	(0.078)	(0.154)	(0.465)	(1.680)
$\text{ATT}_{post_{k=3,4,5}}$	10.721***	0.443***	-1.291***	2.685***	6.575**
	(3.328)	(0.112)	(0.234)	(0.552)	(2.581)

Note: The dependent variable is formal employment over the working-age population according to the 2005 census. Regressions were weighted by the local working-age population in the 2005 census. The CHR refers to commerce, hotels, and restaurants. Primary and secondary include manufacturing, construction, agriculture, and mining. Standard errors are clustered at the municipality level. The coefficients represent percentage point changes. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: PILA 2010-2018 in August.

C Hard discount entry data

To determine the year of entry of hard discount chains into Colombian municipalities, we created a dataset of active HDS in October 2020, including their locations and a proxy for their opening date. The location variable was obtained via web scraping, whereas the opening date comes from the store registration date in the Chambers of Commerce. We then matched the store's location with the date using the store's name. This section describes the process of constructing this match in further detail.

C.1 Web scraping of HDS location

We web-scraped the websites of the three largest chains in October 2020, specifically the sites of their active stores. The sites typically include the store's name, its address (including the municipality and department), and its opening time. We collected information on the first two variables for 2,938 HDS. Importantly, we added the chamber of commerce associated with the municipality for matching purposes.

C.2 Store registration date

We collected data on the universe of establishments registered with the chambers of commerce for the three largest chains as of October 2020. In this dataset, establishments are stores, distribution centers, or stockrooms, both active and inactive. Each table (one per company) contains the establishment name, the chamber of commerce where it was registered, the registration date, and the status (active or inactive). This dataset comprises 3,449 observations.

C.3 Match process

We matched the web-scraped stores with the chambers of commerce dataset on the establishment date of the register using exact, fuzzy, and manual matching.[C.1]

1. We consider two variables when performing exact matching: the store name and the Chamber of Commerce. That means a web-scraped store must match an establishment that shares its name and is registered with the municipality's corresponding chamber of commerce. Around half of the stores (1,453) are matched using this method.

2. We executed two rounds of fuzzy matching, using the *Jaccard* index as a string distance measure, with $q = 3$ as the q-gram size.[C.2] For the first round, we discarded the matches with an index higher than 0.8 (43 stores). After manually revising the matches and confirming the Chamber of Commerce match, we discarded 431 stores. At the end of the first round, 2,484 stores (around 85% of the active stores) are matched.

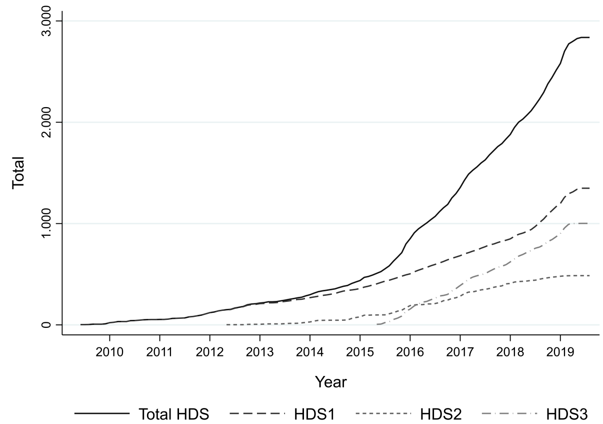
3. We repeated the fuzzy matching using the sample of unmatched web-scraped stores and unmatched registered establishments. In this second round, we did not discard matches based on the Jaccard Index or the coincidence of the chamber of commerce. After a manual review, we additionally matched 259 stores. By the end of this round, 2,743 stores have been matched, representing 93% of the web-scraped stores.

[C.1] Regardless of the matching algorithm, we always matched datasets of the same chain (i.e., web-scraped *Justo y Bueno* stores are always matched with *Justo y Bueno* registered establishments).

[C.2] The Jaccard index is defined as $1 - |X \cap Y|/|X \cup Y|$, where X and Y represent the set of q-grams of size $q = 3$ (subsequences of 3 consecutive characters) in the two strings that are being compared.

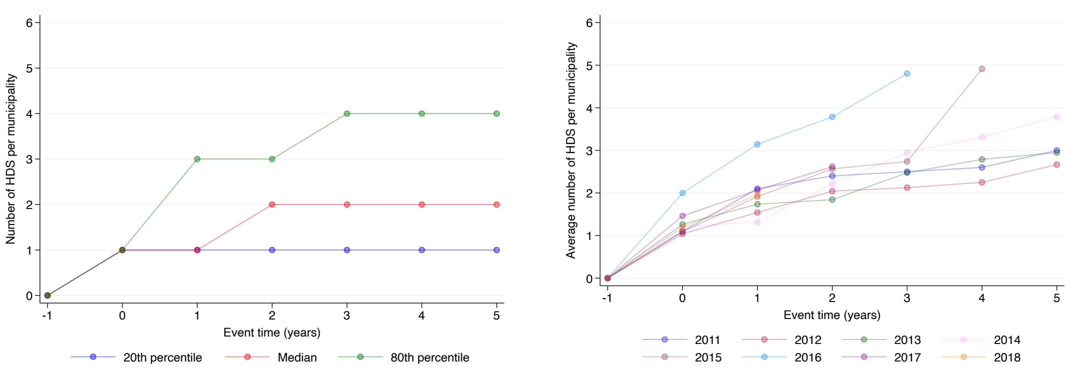
4. Finally, manual matching involves renaming the unmatched registered stores to match the web-scraped stores. Using this method, 105 additional stores are matched. After all the steps, we have a final dataset of 2,847 matched stores.

Figure C.1: Number of HDS over time



Note: This figure shows the total number of hard discount stores from the three main chains operating in Colombia between 2010 and 2019. Source: Authors' calculations using public location data obtained from the hard discounters' websites.

Figure C.2: Number of HDS by event time per municipality



(a) Statistics of all cohorts

(b) Average by cohort

Note: We restrict the event time after five years. Source: Authors' calculations using public location data from hard discounters' websites.

C.4 Measurement error of HDS entry data

We use an externally constructed dataset from a leading supermarket to assess the accuracy of our internally collected opening dates, focusing specifically on the opening year. We compare the internal and external measures at both the municipality and store levels. A caveat is that the external measure is incomplete before 2017, while 40% of the stores in our sample opened before 2017.

Analysis at the municipality level: We compare the year in which a chain first arrived in a municipality using the internal and external measures, restricting the analysis to municipalities in which a chain opened between 2017 and 2020 according to the internal measure. The year of arrival coincides across the two measures in 88% of municipalities for HDS2 and 47% for HDS1. More broadly, the difference between the two measures is one year or less in 96% of municipalities for HDS2 and 89% for HDS1. This exercise validates the accuracy of our opening entry date, as it differs by more than one year in a few cases. The distribution of the internal measure relative to the external measure is:

- Same year: 88% for HDS2 and 47% for HDS1.
- One year earlier: 6.4% for HDS2 and 42.2% for HDS1.
- One year later: 1.6% for HDS2 and 0.7% for HDS1.

Analysis at the store level: We match individual stores across the internal and external measures and compare their opening years. Because store names may differ across sources, we use both exact and fuzzy matching algorithms.

Using an exact match based on store name, municipality, and chain, 62.5% of matched stores have the same opening year across the two measures, while 91.77% differ by one year or less. Using a fuzzy match based on store name and chain, with fuzzy matching on the store name, 59% of matched stores have the same opening year, while approximately 85% differ by one year or less. Again, we validate the accuracy of the entry data even at the store level.

C.5 Determinants of HDS entry

Hard discount chains decide entry based on constant municipality characteristics, such as potential market size and proximity to major capital cities. Table C.1 presents cross-sectional correlations between a municipality's probability of receiving HDS and its population (a proxy for market size), the share of rural population (a proxy for economic development), driving time to Bogotá and Medellín (where the largest logistical centers are based), and the formal employment share in the service industry (a proxy for sectoral composition). We estimate this probability by chain and for all chains combined. Population, the share of rural population, and driving time to Bogotá and Medellín play a significant role in determining store openings (R^2 of 0.538), with population size as the main factor, while the formal tertiary employment share shows the weakest correlation.

Table C.2 shows the correlation between these same characteristics and the year of discounters' arrival relative to 2009, excluding never-treated municipalities. In models with a single variable, which also include department-level fixed effects, log population, driving time to Bogotá, and the share of rural population display the most statistically significant correlations. When all are added to the model, only population and driving time to Bogotá remain statistically significant, at the 1% and 10% level, respectively. As in the models predicting the likelihood of a municipality receiving a hard discount store, the R^2 of these models is also high (0.573 in the regression including all variables), highlighting the importance of these time-invariant characteristics for predicting the timing of entry.

A potential concern for our *timing* assumption, however, is that hard discount chains might instead decide when to open their stores based on local employment dynamics or tax collection, which would violate the parallel-trends assumption underlying our design. We have already tested this empirically with the dynamic specification for each outcome separately and found flat pre-trends, although we have not tested across multiple outcomes simultaneously. For this exercise, we run cohort-specific regressions of the probability that a store opens in a given municipality on the fixed characteristics above, together with joint dynamic characteristics: the annual growth in formal employment, wages, and tax revenues relative to the previous year, all measured using administrative data. Table C.3 reports the correlations of these dynamic covariates for the not-yet-treated municipalities (the time-invariant characteristics are included but not displayed, for simplicity). With the exception of the formal wage growth coefficient in the 2015 regression, none of the other coefficients are statistically significant. Trends in formal employment, wages, and tax collection therefore do not appear to play a key role in the timing of entry.

Table C.1: Treatment selection by hard discount chain

	HDS1		HDS2		HDS3		Any chain	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Total population (in logs)	0.174***	0.193***	0.107***	0.102***	0.164***	0.176***	0.187***	0.200***
	(0.013)	(0.011)	(0.023)	(0.025)	(0.020)	(0.016)	(0.018)	(0.014)
Driving time to Bogota in hours	-0.014***	-0.006	-0.010	-0.007	-0.017***	-0.013**	-0.010***	-0.005
	(0.005)	(0.009)	(0.007)	(0.007)	(0.004)	(0.006)	(0.003)	(0.008)
Driving time to Medellin in hours	-0.027***	-0.030***	0.000	0.000	0.004	-0.009	-0.019***	-0.029***
	(0.005)	(0.007)	(0.012)	(0.009)	(0.005)	(0.008)	(0.003)	(0.006)
Share of rural population	-0.005***	-0.005***	-0.005***	-0.004***	-0.005***	-0.006***	-0.007***	-0.006***
	(0.001)	(0.000)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.000)
Share of formal tertiary employment	-0.001**	-0.000	0.000	0.001	-0.001	0.000	-0.002**	-0.001
	(0.000)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Municipalities	1,038	1,037	1,038	1,037	1,038	1,037	1,038	1,037
R-squared	.472	.516	.29	.454	.398	.502	.494	.538
Dep Var Mean	.282	.282	.168	.167	.237	.236	.387	.387
Dep Var SD	.45	.45	.374	.373	.425	.425	.487	.487
Department F.E	NO	YES	NO	YES	NO	YES	NO	YES

Note: This table presents the estimation results of four cross-sectional linear probability models whose dependent variables indicate whether a municipality had received a store from a particular hard discount chain by 2019 (in Columns 1 to 6), or from any chain in Columns 7 and 8. The municipal population and the share of the rural population were measured in 2005, the share of formal tertiary employment in 2010, and the driving times in 2024. Even columns also incorporate department dummies. Standard errors, clustered at the department level, are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.2: Correlates with the relative year of hard-discount entry

	(1)	(2)	(3)	(4)	(5)	(6)
Total population (in logs)	-0.688***					-0.645***
	(0.200)					(0.222)
Driving time to Bogota in hours		0.515***				0.331*
		(0.176)				(0.169)
Driving time to Medellin in hours			0.582			0.378
			(0.351)			(0.337)
Share of rural population				0.031***		0.006
				(0.007)		(0.004)
Share of formal tertiary employment					-0.013*	-0.001
					(0.007)	(0.007)
Constant	14.374***	2.730*	2.607	5.983***	8.173***	7.872***
	(2.066)	(1.528)	(2.769)	(0.297)	(0.517)	(1.430)
Observations	406	399	399	406	406	399
R-squared	.455	.464	.476	.449	.389	.573
Department FE	YES	YES	YES	YES	YES	YES

Note: This table presents the estimates of 6 cross-sectional models whose dependent variable indicates the year of entry of hard discounters to a municipality relative to 2009 (Year of Entry – 2009). Regressions include fixed effects at the department level, plus the variable(s) indicated in the left column. The overall mean of the dependent variable is 7.25, with a standard deviation of 2.49. Standard errors clustered at the department level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.3: Treatment selection by year (excluding never treated)

	(1) 2011	(2) 2012	(3) 2013	(4) 2014	(5) 2015	(6) 2016	(7) 2017	(8) 2018
Dynamic characteristics:								
Formal employment growth	0.019	0.041	-0.070	-0.079	-0.116	0.007	0.087	-0.216
	(0.014)	(0.045)	(0.090)	(0.074)	(0.105)	(0.106)	(0.131)	(0.254)
Formal wages growth	0.052	-0.033	0.085	0.178	0.461*	0.079	0.104	-0.766
	(0.077)	(0.042)	(0.107)	(0.175)	(0.264)	(0.157)	(0.312)	(0.505)
Income tax growth	0.005	-0.009	-0.025	0.026	0.032	-0.015	-0.007	0.248
	(0.023)	(0.025)	(0.024)	(0.021)	(0.030)	(0.080)	(0.105)	(0.153)
# of total municipalities	389	378	352	333	313	287	225	143
# of muni. treated that year	11	27	19	20	26	62	82	65
R-squared	.0936	.144	.0702	.0952	.068	.136	.148	.159
Time-invariant variables	YES	YES	YES	YES	YES	YES	YES	YES

Note: This table presents the estimation results of yearly cross-sectional linear probability models in which the dependent variable indicates whether a municipality received its first hard discount store in a given year, with later-treated municipalities as the control group. The municipal population and the share of the rural population were measured in 2005, the share of formal tertiary employment in 2010, and the driving times in 2024. Dynamic characteristics are calculated as the annual difference divided by the previous year's value. Standard errors, clustered at the department level, are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D Wage-premiums of hard discounters

To better understand how hard discount chains differ from other formal retail firms in their wage-setting policies, we estimate the canonical model of Abowd et al. (1999) to isolate firm-specific wage determinants. Using the output from this estimation, we answer how much HDS pays its workers, on average, relative to other firms after controlling for worker characteristics. This is relevant because recent evidence from developing countries (Bassier, 2023; Delgado-Prieto, 2024b; Gerard et al., 2021; Pérez Pérez and Nuño-Ledesma, 2024) suggests that the role of firm pay policies in explaining wage inequality may be greater in developing than in developed countries. We estimate our AKM model using the standard log-linear specification:

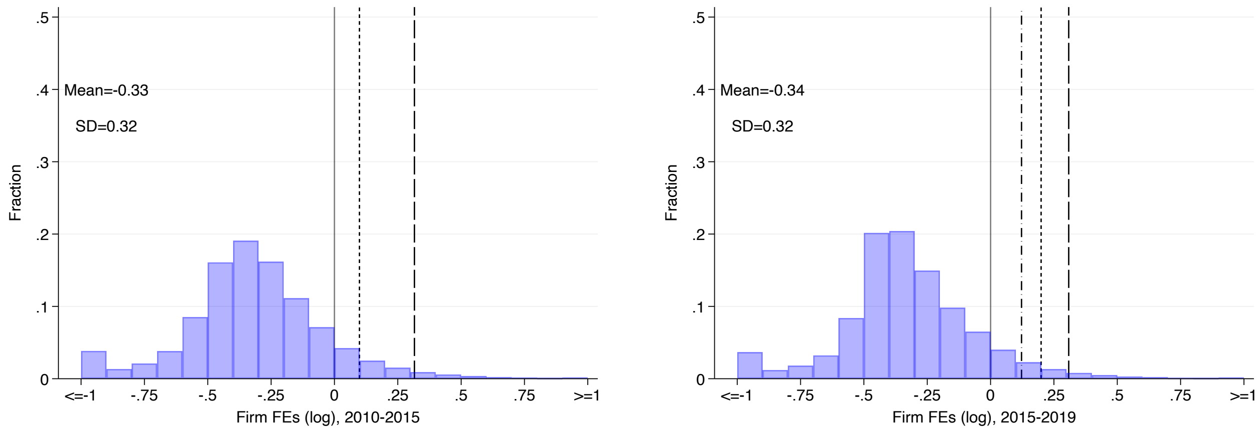
$$log(wages_{it}) = \alpha_i + \psi_{j(i,t)} + X_{it}'\xi + \varepsilon_{it} \tag{D.1}$$

Here, $log(wages_{it})$ denotes the log of monthly wages. They are a function of an additive linear combination of unobserved worker Fixed Effects (FEs) α_i and unobserved firm FEs ψ_j. To identify the latter component, we restrict to the largest set of workers and firms connected by workers' mobility, eliminating around 3% of the initial sample of workers aged 20 to 60. The $j(i,t)$ refers to the firm j of worker i in period t, and the vector X_{it} consists of time-varying controls, which are age squared and its cube (after normalization at age 40) and year FEs. We divided the estimation period into two segments: from 2010 to 2015, when the first HDS opened and primarily employed skilled workers, and from 2015 to 2019, when they expanded extensively across the country and increasingly employed blue-collar workers. In this exercise, all the firm FEs are relative to the largest retail store in the country, so on average, they are negative.

Figure D.1 shows that the three hard discount chains, in dashed lines, pay consistently higher premiums to all their workers than the country's largest retail store or supermarket. They are located at the higher end of the wage premium distribution, indicating that, after controlling for worker characteristics, these firms contribute positively to the wages of all their workers. In both estimation periods, we find a positive contribution from hard discount chains. This is consistent with evidence from Alfaro-Urena et al. (2022), which shows that multinationals pay a positive wage premium to Costa Rican workers. To read the figure more intuitively, when a worker moves from the reference firm at zero to one of the discount chains, they would experience an average positive gain in log wages of approximately the value of the dashed line. Lastly, note that the firm's pay premiums are measured at the national level for all locations and workers, including managers and blue-collar workers.[D.1]

[D.1] We do not use the leave-out method proposed by Kline et al. (2020) for the estimation, as it yields unbiased variance and covariance moments of the wage decomposition, while we use the vector of estimated level parameters.

Figure D.1: Distribution of Firm FEs



(a) 2010–2015

(b) 2015–2019

Note: The dashed lines are the HDS chains (from 2010 to 2015, there are only two), while the retail firm used as reference is located at the zero line. For confidentiality reasons, we do not disclose which line belongs to which HDS. For the estimation sample, we exclude workers with non-positive wages, those with fewer than 30 employment days per month, and employees aged 20 to 60 years. Additionally, we retain the highest-wage contract for workers who make multiple contributions to the social security system. Moreover, we eliminate workers and firms that do not belong to the largest connected set of firms and workers, as well as workers who appear only once in the estimation sample. We transform nominal wages into real terms using the monthly CPI from DANE (base year 2018). Source: PILA August 2010–August 2019.

E Robustness Checks

E.1 Baseline covariates and potential confounders

We assess robustness to baseline covariates motivated by the selection patterns discussed in Section 2: population, driving time to Bogotá and Medellín, and the share of rurality are correlated with the probability of receiving a store, both overall and in a given year. Our baseline estimates already account for population by weighting regressions with the working-age population from the 2005 census. Figures E.2 to E.4 and accompanying tables report the estimates when we adjust our estimator to control for the remaining covariates.[E.1]

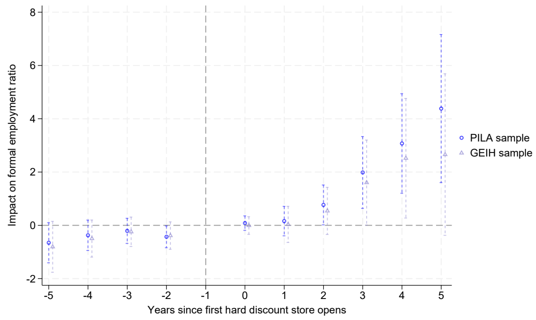
[E.1] We opt to incorporate these covariates using the outcome regression DiD estimator based on ordinary least squares instead of other estimators such as the doubly robust estimator, whose propensity-score methods rely on a common support assumption that is unlikely to hold in our case (Figure B.1).

Table E.1: Average *C&S* estimates of formal employment ratios by sector controlling for baseline informality

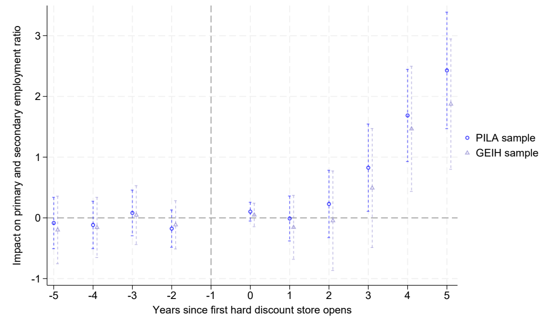
	(1) Total	(2) Retail	(3) CHR without retail	(4) Primary and secondary	(5) Services without CHR
Panel A: PILA-Sourced Outcomes					
ATT$_{pre}$	-0.476	-0.019	0.002	-0.105	-0.151
	(0.364)	(0.035)	(0.120)	(0.271)	(0.280)
ATT$_{post}$	1.224*	0.067	-0.241	0.632	0.598
	(0.692)	(0.062)	(0.225)	(0.419)	(0.641)
ATT$_{post_{k=3,4,5}}$	2.249*	0.106	-0.357	1.314**	0.749
	(1.199)	(0.090)	(0.306)	(0.573)	(0.951)
Panel B: GEIH-Sourced Outcomes					
ATT$_{pre}$	0.322	-0.064	-0.063	0.263	0.186
	(0.681)	(0.179)	(0.159)	(0.336)	(0.516)
ATT$_{post}$	3.040***	0.577***	-0.047	1.776***	0.735
	(0.924)	(0.220)	(0.203)	(0.455)	(0.520)
ATT$_{post_{k=3,4,5}}$	4.686***	0.797***	0.241	2.511***	1.138
	(1.255)	(0.286)	(0.287)	(0.618)	(0.749)

Note: This table reports the robustness of the formal employment estimates to the inclusion of baseline informality as a covariate. The dependent variable is formal employment over the working-age population according to the 2005 census. The CHR refers to commerce, hotels, and restaurants. Primary and secondary are manufacturing, construction, agriculture, and mining. We define baseline informality as informal employment in 2010 (sourced from survey data) over the working-age population according to the 2005 census. We incorporate this covariate into the model using the outcome regression DiD estimator based on ordinary least squares. Regressions were weighted with the working-age population of the 2005 census, and standard errors were clustered at the municipality level. The coefficients represent percentage point changes. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: PILA 2010-2018 in August (Panel A), GEIH 2010-2018 (Panel B).

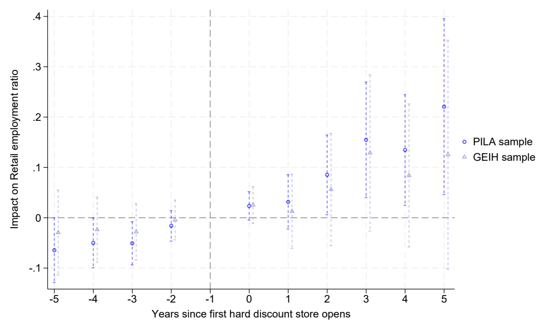
Figure E.1: Event study estimates for administrative-data formal employment ratios using municipalities in survey data (GEIH)



(a) Total employment



(b) Primary & secondary



(c) Retail

Note: The dependent variable is formal employment in a given sector using administrative data, divided by the working-age population in the 2005 census. We restrict the sample to municipalities available in the survey data (GEIH). Regressions are weighted by the local working-age population in the 2005 census, and standard errors are clustered at the municipality level. 90% confidence intervals. The coefficients represent percentage point changes. Source: PILA 2010-2018.

Most of the main results remain robust to the inclusion of covariates. For formal employment based on administrative data, Figure E.2a shows that the estimates remain positive and statistically significant, especially in the later years. Regarding the industry-level heterogeneous effects, we still find an upward trend in the primary and secondary sectors, but the estimates become less precise (see Figure E.2b). Relatedly, Figure E.3b shows that, when measuring formal employment in these sectors using survey data, the estimates are robust to controls, suggesting that these stores continue to have a positive impact even after adjusting for covariates. Other labor market outcomes do not show substantial changes after adjustment for covariates, except for formal employment estimates from survey data. Figure E.3a shows that when adding controls, the post-treatment estimates reduce their magnitude and become more imprecise. However, they remain positive and show an upward trend over time.[E.2] Regarding employment, unemployment, and inactivity rates, Table E.2 shows that the estimated impacts on these outcomes become larger and more significant. Next, the effects on informal employment (Table E.3), labor income, and working hours (Table E.4) remain non-significant. Regarding the tax outcomes, Figures E.4a and E.4b show how the dynamic impact estimates differ when including one control at a time versus all controls. Columns 2 and 4 of Table E.5 report the estimated impacts when all covariates are taken into account. Both dependent variables appear robust to the inclusion of these controls, with the coefficients increasing in magnitude.

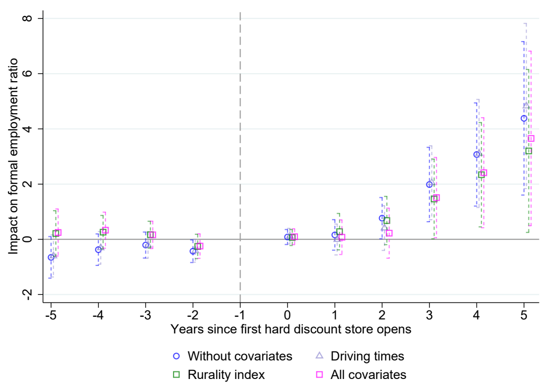
An additional confounding factor might be migration, both internal and from Venezuela. This is unlikely to drive our results, as migration flows would need to be systematically aligned with store entry timing. Nonetheless, we assess whether the Venezuelan exodus to Colombia affects our findings. Note that we already exclude capital cities from the analysis (where most migrants settle and internal migration is most concentrated) and instead focus on small urban cities. Still, we directly account for the immigration shock, using data from the 2018 census. We define this as the ratio of employed Venezuelan migrants in a municipality to the total employed population aged 18 to 64. Panel C of Table E.6 presents the estimates, which remain similar in magnitude and statistical significance to those in the specification without controls (Panel A).[E.3]

Another potential concern is the concentration of effects in metropolitan areas, where the expansion of hard discount chains is easier due to their proximity to capital cities. We have excluded capital cities from our estimation sample to partially address this. However, the estimates can be confounded by municipalities more closely linked to capitals, in ways not captured by the two distance controls. To mitigate this concern, we conducted additional analysis excluding the six largest metropolitan areas. This further exclusion reduces our main sample by 19 municipalities. Panel B of Table E.6 reports that the treatment effects do not vary substantially in magnitude.

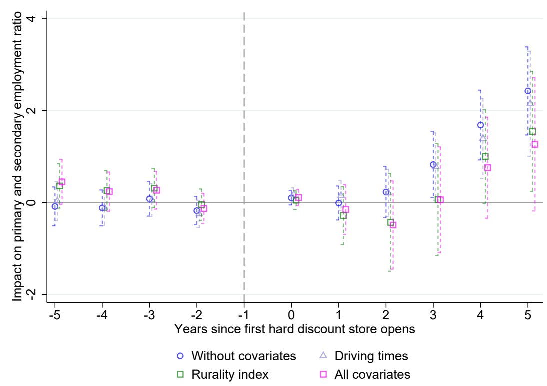
[E.2]For some periods (second and fourth years after the arrival), the coefficients are less precise, but their magnitude does not change, and for others (the third and the fifth years), the magnitudes decrease only when controlling for the distance to Bogotá and Medellín.

[E.3]Besides, internal migration patterns remained stable during our study period. According to Acosta and Gu (2024), five-year migration flows from 2012 to 2019, based on household surveys, consistently show Bogotá as the primary destination. This pattern holds across different definitions (one-year vs. five-year flows).

Figure E.2: Event study estimates for administrative data formal employment ratios including controls



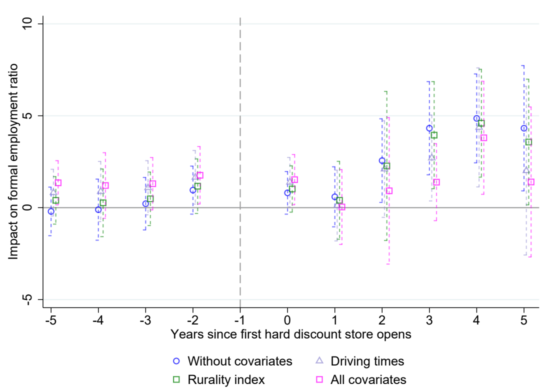
(a) Total Employment



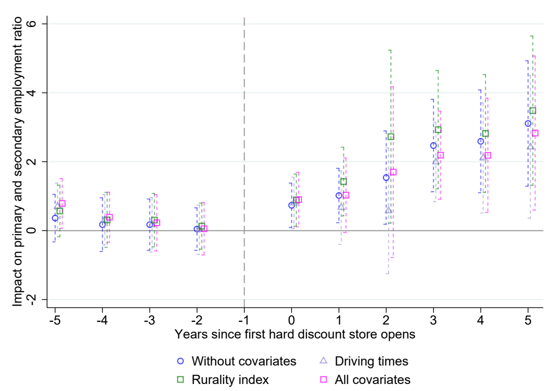
(b) Primary & secondary

Note: The dependent variable is formal employment in a given sector using administrative data, divided by the working-age population in the 2005 census. We control for the share of the rural population measured in 2005 and the driving times to Bogotá and Medellín using the outcome regression DiD estimator based on ordinary least squares. Regressions are weighted by the local working-age population in the 2005 census, and standard errors are clustered at the municipality level. 90% confidence intervals. The coefficients represent percentage point changes. Source: PILA 2010-2018.

Figure E.3: Event study estimates for survey data formal employment ratios including controls



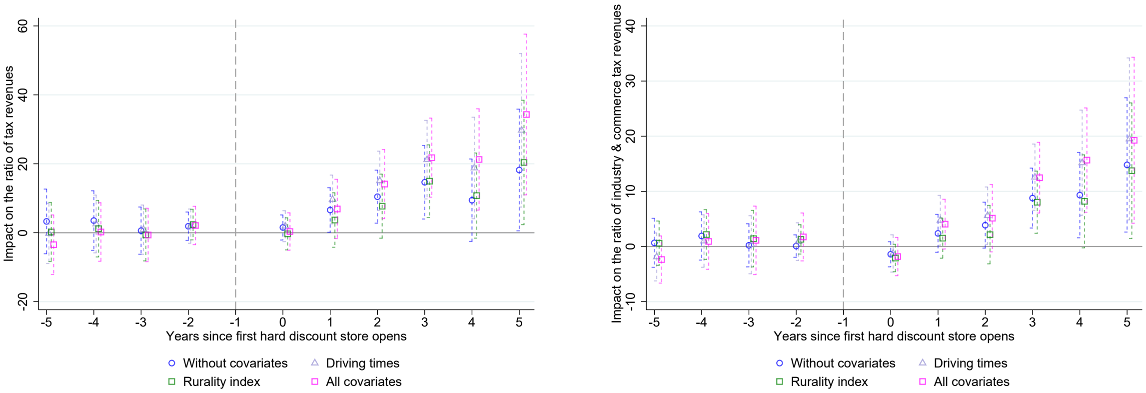
(a) Total Employment



(b) Primary & secondary

Note: The dependent variable is formal employment in a given sector using survey data, divided by the working-age population in the 2005 census. We control for the share of the rural population measured in 2005 and the driving times to Bogotá and Medellín using the outcome regression DiD estimator based on ordinary least squares. Regressions are weighted by the local working-age population in the 2005 census, and standard errors are clustered at the municipality level. 90% confidence intervals. The coefficients represent percentage point changes. Source: GEIH 2010-2018.

Figure E.4: Event study estimates for tax outcomes including controls



(a) Tax revenues

(b) Industry and commerce

Note: The dependent variable in panel (a) is total tax revenue divided by all revenues, including taxes and central government transfers. In panel (b), it is industry and commerce tax revenue divided by all revenues. We control for the share of the rural population measured in 2005 and the driving times to Bogotá and Medellín using the outcome regression DiD estimator based on ordinary least squares. Regressions are weighted by municipalities' 2010 revenues, and standard errors are clustered at the municipality level. 90% confidence intervals. The coefficients represent percentage point changes. Source: DNP 2010-2018.

Table E.2: Average *C&S* estimates of labor market indicators using controls

	(1) Employment rate	(2) Unemployment rate	(3) Inactivity rate
Panel A: Baseline			
ATT_{pre}	0.718	-0.477	-0.439
	(0.536)	(0.398)	(0.443)
ATT_{post}	2.286***	-0.990	-1.886**
	(0.849)	(0.628)	(0.728)
$\text{ATT}_{post_{k=3,4,5}}$	3.242***	-1.758**	-2.369**
	(1.191)	(0.884)	(1.058)
Panel B: Including Covariates of Distance to Main Cities and Rurality Index			
ATT_{pre}	0.077	-0.181	0.159
	(0.656)	(0.493)	(0.538)
ATT_{post}	3.234***	-1.539**	-2.694***
	(1.077)	(0.762)	(0.867)
$\text{ATT}_{post_{k=3,4,5}}$	4.611***	-2.247**	-3.765***
	(1.334)	(0.927)	(1.104)

Note: This table reports the robustness to the inclusion of covariates of the labor market rate estimates (Table F.4). The dependent variable in each column is the yearly labor market rate. Panel B controls for the share of rural population and the driving times to Bogotá and Medellín. We incorporate these covariates into the model using the outcome regression DiD estimator based on ordinary least squares. Regressions were weighted by local employment in 2010, and standard errors were clustered at the municipality level. The coefficients represent percentage point changes. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: GEIH 2010-2018.

Table E.3: Average *C&S* estimates of informal employment ratios by sector using controls

	(1) Total	(2) Retail	(3) CHR without retail	(4) Primary and secondary	(5) Services without CHR
Panel A: Baseline					
ATT_{pre}	0.980	0.085	0.413	-0.718	1.199***
	(0.782)	(0.357)	(0.318)	(0.473)	(0.417)
ATT_{post}	-1.105	0.281	-0.223	-0.806	-0.357
	(1.467)	(0.573)	(0.429)	(0.856)	(0.496)
$\text{ATT}_{post_{k=3,4,5}}$	-2.219	0.017	-0.674	-0.441	-1.120*
	(1.729)	(0.756)	(0.569)	(1.054)	(0.627)
Panel B: Including Covariates of Distance to Main Cities and Rurality Index					
ATT_{pre}	-0.290	-0.115	-0.094	-1.001*	0.920*
	(1.007)	(0.455)	(0.424)	(0.534)	(0.519)
ATT_{post}	0.503	0.904	-0.290	-0.244	0.133
	(1.558)	(0.676)	(0.691)	(1.191)	(0.602)
$\text{ATT}_{post_{k=3,4,5}}$	-0.193	0.824	-0.776	0.323	-0.563
	(1.974)	(0.890)	(0.901)	(1.353)	(0.749)

Note: This table reports the robustness to the inclusion of covariates of the informal employment estimates (Table F.10). The dependent variable is informal employment by the given sector relative to the working-age population according to the 2005 census of a given municipality. The CHR refers to commerce, hotels, and restaurants. Primary and secondary sectors include manufacturing, construction, agriculture, and mining. Panel B controls for the share of rural population and the driving times to Bogotá and Medellín. We incorporate these covariates into the model using the outcome regression DiD estimator based on ordinary least squares. Regressions were weighted with the working-age population of the 2005 census, and standard errors were clustered at the municipality level. The coefficients represent percentage point changes. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: GEIH 2010-2018.

Table E.4: Average *C&S* estimates of labor income and working hours using controls

	(1)	(2)	(3)	(4)	(5)	(6)
	Labor income				Working hours	
	Total formal	Total informal	Retail formal	Retail informal	Total formal	Total informal
Panel A: Baseline						
ATT$_{pre}$	0.030	0.002	-0.062	0.007	0.002	0.004
	(0.023)	(0.016)	(0.050)	(0.035)	(0.007)	(0.009)
ATT$_{post}$	-0.022	-0.033	-0.040	-0.078	-0.004	-0.009
	(0.034)	(0.027)	(0.069)	(0.067)	(0.008)	(0.012)
ATT$_{post_{k=3,4,5}}$	-0.041	-0.027	-0.032	-0.132	-0.007	-0.014
	(0.045)	(0.037)	(0.102)	(0.106)	(0.010)	(0.017)
Panel B: Including Covariates of Distance to Main Cities and Rurality Index						
ATT$_{pre}$	0.036	-0.025	-0.087	0.002	0.005	0.002
	(0.025)	(0.020)	(0.054)	(0.038)	(0.007)	(0.009)
ATT$_{post}$	-0.061*	-0.016	-0.058	-0.106	0.013	0.015
	(0.036)	(0.033)	(0.094)	(0.072)	(0.013)	(0.014)
ATT$_{post_{k=3,4,5}}$	-0.082	0.002	0.001	-0.138	0.007	0.005
	(0.054)	(0.041)	(0.143)	(0.112)	(0.015)	(0.020)

Note: This table reports the robustness to the inclusion of covariates of wages and working hours estimates using survey data (Figures 8a and 8c). The dependent variables are the logarithm of sectoral labor incomes in Columns 1 to 4, and of working hours in Columns 5 and 6. Panel B controls for the share of rural population and the driving times to Bogotá and Medellín. We incorporate these covariates into the model using the outcome regression DiD estimator based on ordinary least squares. Regressions were weighted with the working-age population of the 2005 census, and standard errors were clustered at the municipality level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: GEIH 2010-2018.

Table E.5: Average *C&S* estimates of tax ratios using controls

	(1)	(2)	(3)	(4)
	Ratio of tax revenues		Ratio of industry & commerce tax revenues	
	Without controls	With controls	Without controls	With controls
ATT$_{pre}$	2.321	-0.453	0.718	0.354
	(4.003)	(4.244)	(1.927)	(2.778)
ATT$_{post}$	10.138**	16.447***	6.291**	9.127**
	(4.740)	(5.779)	(2.939)	(3.616)
ATT$_{post_{k=3,4,5}}$	14.091*	25.776***	10.967**	15.799**
	(7.374)	(9.010)	(4.741)	(5.673)

Note: This table reports the robustness of the two main tax outcomes (total tax revenues and revenues from the industry and commerce tax) to the inclusion of covariates. The dependent variable in Columns 1 and 2 is the ratio of taxes over all revenues (including taxes and central government transfers), and it corresponds to the ratio of industry and commerce tax revenues over all revenues in Columns 3 and 4. While estimates in Columns 1 and 3 do not include covariates (they match those in Table F.11), Columns 2 and 4 control for the share of rural population and the driving times to Bogotá and Medellín. We incorporate these covariates into the model using the outcome regression DiD estimator based on ordinary least squares. Regressions were weighted by each municipality's 2010 revenues, and standard errors were clustered at the municipality level. The coefficients represent percentage point changes. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: DNP 2010-2018.

Table E.6: Average *C&S* estimates of formal employment ratios excluding metropolitan areas and accounting for the immigration shock

	(1) Total	(2) Retail	(3) CHR without retail	(4) Primary and secondary	(5) Services without CHR
Panel A: Baseline					
ATT_{pre}	-0.417	-0.045*	0.030	-0.074	-0.100
	(0.294)	(0.025)	(0.090)	(0.205)	(0.211)
ATT_{post}	1.742***	0.108**	-0.290*	0.877***	0.670
	(0.608)	(0.044)	(0.158)	(0.296)	(0.474)
$\text{ATT}_{post_{k=3,4,5}}$	3.147***	0.170**	-0.397*	1.647***	1.013
	(1.049)	(0.066)	(0.228)	(0.426)	(0.756)
Panel B: Excluding metro-area municipalities					
ATT_{pre}	-0.547*	-0.055**	0.069	-0.144	-0.194
	(0.318)	(0.027)	(0.101)	(0.229)	(0.240)
ATT_{post}	1.732**	0.133***	-0.293*	0.857***	0.591
	(0.676)	(0.045)	(0.168)	(0.309)	(0.524)
$\text{ATT}_{post_{k=3,4,5}}$	3.098***	0.206***	-0.405	1.636***	0.871
	(1.154)	(0.066)	(0.250)	(0.442)	(0.844)
Panel C: Including the Venezuelan immigration shock					
ATT_{pre}	-0.444	-0.048*	0.033	-0.099	-0.140
	(0.291)	(0.025)	(0.091)	(0.204)	(0.212)
ATT_{post}	1.753***	0.114***	-0.288*	0.889***	0.664
	(0.609)	(0.042)	(0.160)	(0.303)	(0.490)
$\text{ATT}_{post_{k=3,4,5}}$	3.145***	0.175***	-0.393*	1.657***	1.016
	(1.050)	(0.065)	(0.229)	(0.431)	(0.766)

Note: This table reports the robustness of the formal employment estimates to excluding municipalities belonging to a metropolitan area and to including a covariate for the Venezuelan migration shock. The baseline estimates correspond to those from administrative data (Table F.6). CHR refers to commerce, hotels, and restaurants. The primary and secondary sectors include manufacturing, construction, agriculture, and mining. Panel B excludes 19 municipalities classified by DANE as part of metropolitan areas, and Panel C controls for the share of migrants using the outcome regression DiD estimator based on ordinary least squares. We construct the share of migrants using the 2018 census as the ratio of employed migrants from Venezuela in a municipality to the total employed population aged 18 to 64. Regressions are weighted by the working-age population in the 2005 census, and standard errors are clustered at the municipality level. The coefficients represent percentage point changes. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: PILA 2010-2018.

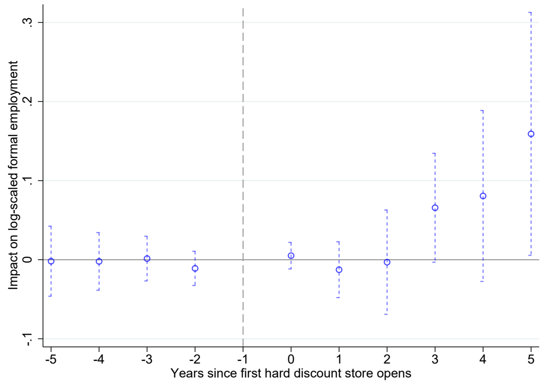
E.2 Regression weights and log transformation

We evaluate robustness to weighting, functional form, and outliers by excluding regression weights, using the logarithm of formal employment and tax collection as dependent variables, and winsorizing observations in the upper and lower tails of our preferred ratio-like outcomes. Our preferred estimation uses population weights to account for potential market size and to focus on the effect on the average individual, rather than on the average region. It also uses ratios of outcomes to mitigate potential concerns arising from differences in pre-treatment outcome distributions between treated and not-yet-treated municipalities. In this subsection, we show that our main results remain largely unchanged when we opt for unweighted models or use a log transformation of the outcomes. Panel B of Table E.7 reports the estimated impacts on formal employment using administrative data with an unweighted specification. Compared to the weighted estimates in Panel A, the unweighted estimates for total formal employment remain positive and statistically significant, with similar results for employment in the primary and secondary sectors. On the other hand, the estimated impacts on retail employment, as well as on hotels and restaurants, become statistically significant. Finally, we still find inconclusive impacts on service employment.

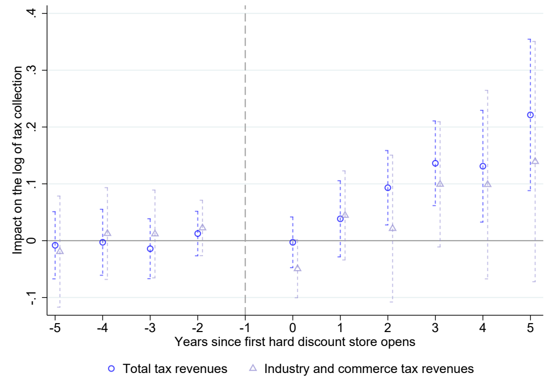
Regarding the log transformation, Table E.8 compares the estimates on total formal employment using controls when the dependent variable is the ratio (Column 1) to when it is in logs (Column 2). Figure E.5a exhibits the dynamic effects. Despite the coefficients becoming more imprecise, the effect remains positive and increases over time: on average, formal employment grows by 4.9% after a municipality receives HDS, and by around 10% after the third year. Robustness is stronger for the results about taxes, as shown in Table E.9 and Figure E.5b. Columns 2 and 5 of the table display the estimated impacts when using the log of total and industry and commerce tax revenues, respectively. For total tax revenues, the coefficients are positive and significant: tax collection in the municipality increases by 10.3% on average after stores' arrival. The effects remain positive and show an upward trend for industry and commerce tax revenues, although they lose statistical significance.

One reason for the loss of statistical significance in estimates using log-transformed outcomes could be the presence of outliers in the preferred ratio specifications. These could be due to measurement or reporting issues, which most likely affect the administrative formal employment data, or to a sudden spike in formal employees in a handful of treated municipalities. To rule out this hypothesis, we check whether our estimates remain statistically significant when applying a 5% winsorization to the ratio-dependent variables. We used the 5th and 95th percentiles of their overall distribution for this exercise. Column 3 of Table E.8 shows the results for formal employment, and Columns 3 and 6 of Table E.9 show the results for tax collection. Although reduced in magnitude, formal employment results maintain the statistical significance of the ratio without winsorization. Estimates for total tax revenue remain largely unchanged, while industry and commerce tax revenues are the least robust: the estimated impact is still positive and growing with time, but it is only significant at 10% for the years 3 to 5 after hard-discount arrival.

Figure E.5: Event study estimates for formal employment and main taxation outcomes in logs



(a) Total formal employment



(b) Taxation outcomes

Note: The dependent variable in panel (a) is the natural logarithm of total formal employment using administrative data. In panel (b), the dependent variables are the natural logarithms of total tax revenues and industry and commerce tax revenues. The regression in panel (a) is weighted by the working-age population of the 2005 census, and those in panel (b) are weighted by municipalities' 2010 revenues. All regressions control for the share of the rural population and driving times to Bogotá and Medellín using the outcome regression DiD estimator based on ordinary least squares. Standard errors are clustered at the municipal level. 90% confidence intervals. Sources: PILA 2010-2018 and DNP 2010-2018.

Table E.7: Average *C&S* estimates of formal employment ratios by sector excluding weights

	(1) Total	(2) Retail	(3) CHR without retail	(4) Primary and secondary	(5) Services without CHR
Panel A: Weighted (Baseline)					
ATT_{pre}	-0.417	-0.045*	0.030	-0.074	-0.100
	(0.294)	(0.025)	(0.090)	(0.205)	(0.211)
ATT_{post}	1.742***	0.108**	-0.290*	0.877***	0.670
	(0.608)	(0.044)	(0.158)	(0.296)	(0.474)
$ATT_{post_{k=3,4,5}}$	3.147***	0.170**	-0.397*	1.647***	1.013
	(1.049)	(0.066)	(0.228)	(0.426)	(0.756)
Panel B: Unweighted					
ATT_{pre}	-0.099	-0.073***	0.057	0.085	0.074
	(0.299)	(0.019)	(0.078)	(0.160)	(0.212)
ATT_{post}	1.901***	0.115***	-0.359***	0.930***	0.469
	(0.643)	(0.035)	(0.126)	(0.275)	(0.384)
$ATT_{post_{k=3,4,5}}$	3.499***	0.185***	-0.378**	1.743***	0.772
	(1.081)	(0.054)	(0.189)	(0.425)	(0.642)

Note: This table reports the robustness of the formal employment estimates when we do not weight them. Panel A reports the results from the regressions that use administrative data and are weighted with the 2005 census working-age population (see Table F.6), while estimates in Panel B do not include weights. The dependent variable is formal employment in the given sector relative to the working-age population, as reported in the 2005 census. The CHR refers to commerce, hotels, and restaurants. Primary and secondary sectors include manufacturing, construction, agriculture, and mining. Standard errors clustered at the municipality level. The coefficients represent percentage point changes. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: PILA 2010-2018.

Table E.8: Average *C&S* estimates of formal employment ratio using outcomes in logs and winsorized

	(1) Formal employment ratio	(2) Logarithm of formal employment	(3) Winsorized formal employment ratio
ATT_{pre}	0.124	-0.003	0.070
	(0.329)	(0.018)	(0.261)
ATT_{post}	1.329*	0.049	0.666*
	(0.678)	(0.038)	(0.394)
$\text{ATT}_{post_{k=3,4,5}}$	2.526**	0.102	1.452**
	(1.178)	(0.062)	(0.635)

Note: This table reports the robustness of the total formal employment estimates to using a log-transformation on the dependent variable and a winsorized version of the preferred ratio-interpreted outcome. The dependent variable in Column 1 is formal employment from administrative data, relative to the working-age population according to the 2005 census; in Column 2, it is the natural logarithm of formal employment; and in Column 3, it is a 5% winsorization of the ratio. All regressions were weighted by the working-age population of the 2005 census and adjusted for the share of rural population and driving times to Bogotá and Medellín using the outcome regression DiD estimator based on ordinary least squares. Standard errors clustered at the municipality level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: PILA 2010-2018.

Table E.9: Average *C&S* estimates of tax ratios using outcomes in logs and winsorized

	(1)	(2)	(3)	(4)	(5)	(6)
		Tax revenues			Industry & commerce tax revenues	
	Ratio	Log-scaled	Winsorized	Ratio	Log-scaled	Winsorized
ATT_{pre}	-0.453	-0.003	0.993	0.354	0.007	-1.003
	(4.244)	(0.028)	(3.802)	(2.778)	(0.042)	(1.287)
ATT_{post}	16.447***	0.103***	14.261***	9.127**	0.059	2.955
	(5.779)	(0.038)	(4.809)	(3.616)	(0.062)	(2.206)
$\text{ATT}_{post_{k=3,4,5}}$	25.776***	0.163***	22.332***	15.799***	0.112	5.201*
	(9.010)	(0.056)	(7.113)	(5.673)	(0.090)	(3.126)

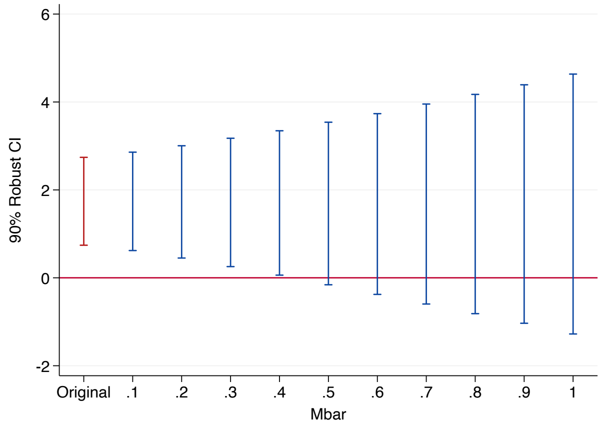
Note: This table reports the robustness of the two main tax outcomes (total tax revenues and revenues from the industry and commerce tax) to using a log-transformation and a winsorization of the preferred ratios. The dependent variables in Columns 1 and 4 are total tax revenues and industry and commerce tax revenues, respectively, each divided by total revenues (taxes and central government transfers). Columns 2 and 5 have, as outcomes, the natural logarithms of total and industry and commerce tax revenues, respectively. The dependent variables in Columns 3 and 6 are the 5% winsorization of the ratios in Columns 1 and 4. All regressions were weighted using municipalities' revenues in 2010 and adjusted for the share of rural population and driving times to Bogotá and Medellín using the outcome regression DiD estimator based on ordinary least squares. Standard errors clustered at the municipality level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: DNP 2010-2018.

E.3 Placebo Tests and Sensitivity to Pre-Trends

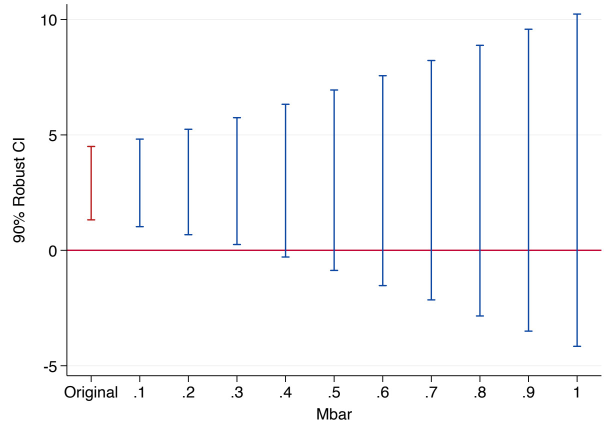
We conduct Fisher randomization tests (FRT) by permuting treatment timing (the year HDS enters each municipality). Since our baseline estimator computes p-values using standard errors clustered at the municipal level, we follow the procedure proposed by Roth and Sant'Anna (2023), which is designed for staggered adoption settings where treatment timing is approximately random and is compatible with the C&S estimator. Table E.10 presents the FRT-based p-values for the event study estimates of our two main formal employment outcomes: total formal employment (columns 1-3) and formal employment in the primary and secondary sectors (columns 4-6). The treatment effect estimates correspond to panel B of Table E.7, as the Roth and Sant'Anna (2023) method does not support the inclusion of regression weights. For each outcome, we report three p-values: the standard z-score-based value used in C&S (relying on clustered standard errors), and two FRT-based p-values (one using covariate-adjusted standard errors and the other using Neyman standard errors). Across all specifications, the p-values are very similar, and the statistical significance of the results remains unchanged when we randomize treatment timing.

Finally, we relax the assumption of parallel pre-trends and perform a sensitivity analysis of the average post-treatment coefficients, following Rambachan and Roth (2023). A potential concern in this setting is that unobserved time-varying shocks at the municipal level would have affected early-treated municipalities differently from later ones, even in the absence of HDS. If we assume that these shocks are of similar magnitude before and after treatment, then we can bound the estimates in relative terms with respect to the maximal violation in pre-trends. In that sense, we can obtain breakdown values from which the coefficient is no longer significant. Figure E.6 shows that the average post-treatment coefficient of formal employment is robust up to between 0.3 and 0.4 maximal violations of pre-trends, a moderate margin that rules out only relatively small deviations from parallel trends and does not guarantee robustness to larger violations. The value does not approach violations close to 1, partly because we analyze six periods, which can increase confidence intervals and decrease breakdown values (Rambachan and Roth, 2023).

Figure E.6: Sensitivity analysis of ATT_{Post} for formal employment



(a) Formal employment (Admin)



(b) Formal employment (Survey)

Note: The coefficient is the average of all post-treatment periods. The $Mbar$ refers to how robust the coefficient is to the maximal violation in pre-trends. For instance, $Mbar = 1$ assumes the maximal pre-treatment violation while $Mbar = 0.5$ assumes half the maximal violation. Standard errors are clustered at the municipality level. 90% robust confidence intervals with the conditional least favorable option.

Table E.10: Fisher Randomization Test p-values for event study estimates of formal employment

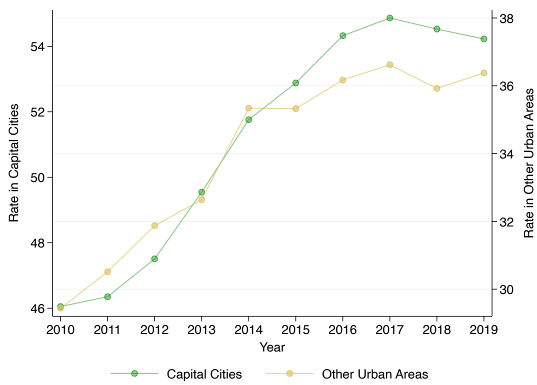
	Total:Clustered	Total:Neyman Fisher	Total:Adjusted Fisher	P&S:Clustered	P&S:Neyman Fisher	P&S:Adjusted Fisher
$ATT_{post_{k=0}}$	0.555	0.566	0.566	0.123	0.121	0.121
$ATT_{post_{k=1}}$	0.530	0.538	0.538	0.947	0.941	0.941
$ATT_{post_{k=2}}$	0.259	0.267	0.267	0.468	0.453	0.453
$ATT_{post_{k=3}}$	0.005	0.004	0.004	0.002	0.000	0.000
$ATT_{post_{k=4}}$	0.001	0.001	0.001	0.000	0.000	0.000
$ATT_{post_{k=5}}$	0.004	0.007	0.007	0.000	0.001	0.001

Note: This table reports robustness of the main formal employment event study estimates to the use of p-values based on randomization tests. The dependent variable is total employment in Columns 1 to 3, and primary & secondary employment in Columns 4 to 6, both according to the administrative data and relative to the working-age population according to the 2005 census. p-values in columns 1 and 4 are those computed using Callaway & Sant'Anna (they rely on standard errors clustered at the municipality level). Those in columns 2, 3, 5, and 6 follow Fisher Randomization Tests (FRT) proposed by Roth and Sant'Anna (2023), whose standard errors are computed using two methods: Neyman in Columns 2 and 5, and covariate-adjusted in Columns 3 and 6. All FRT p-values are based on 5,000 permutations. Source: PILA 2010-2018.

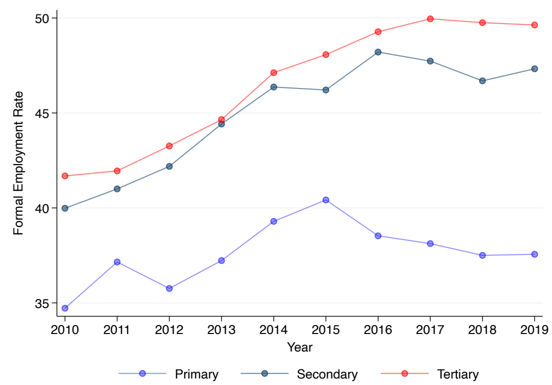
F Supplementary Results

F.1 Figures

Figure F.1: National formal employment rates



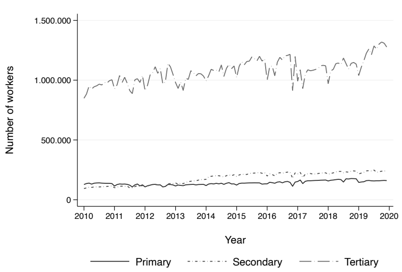
(a) By geographic location



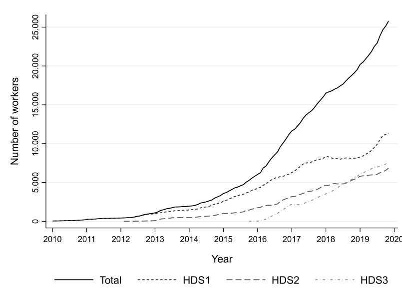
(b) By industry

Note: This figure shows the formality rate by geographic location and by industry groups based on survey data. A worker is considered formal if they contribute to the social security system. We restrict the sample to workers aged 18 to 64 in urban areas. Source: GEIH 2010-2019.

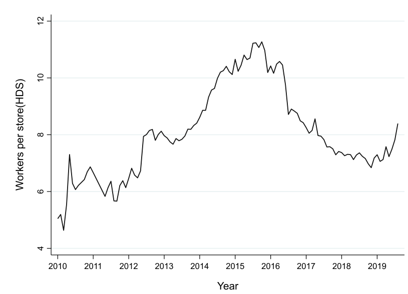
Figure F.2: Descriptive statistics on formal employment at HDS using administrative data



(a) Total FE



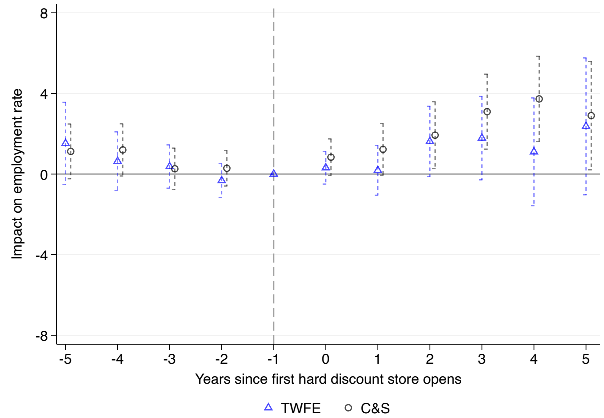
(b) FE in HDS



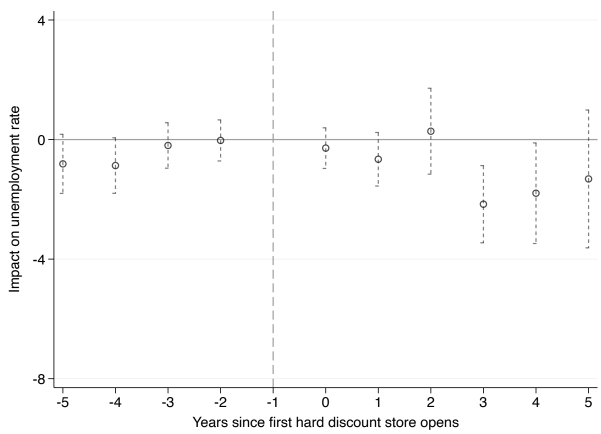
(c) Average full-time FE per HDS

Note: Panel A shows the evolution of total full-time formal employment (FE) in the 372 municipalities included in our sample by industry. Primary refers to employment in agriculture and mining; secondary refers to employment in manufacturing and construction; and tertiary refers to employment in services. Panel B shows the evolution of full-time formal employment by the hard discount chain in the municipalities included in the estimation sample. Panel C reports the average full-time formal employment per store in the municipalities included in the estimation sample. Source: Authors' calculations based on PILA.

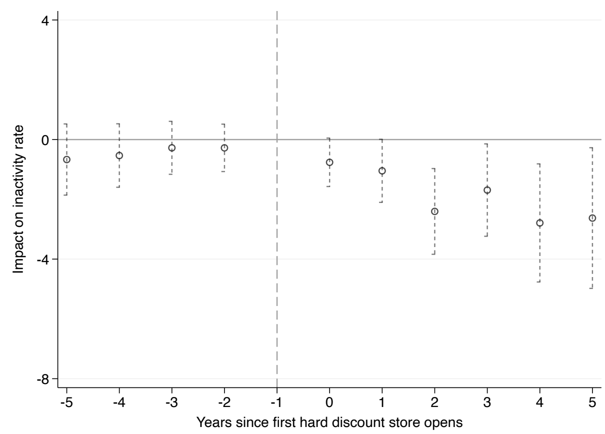
Figure F.3: Event study estimates of labor market rates



(a) Employment



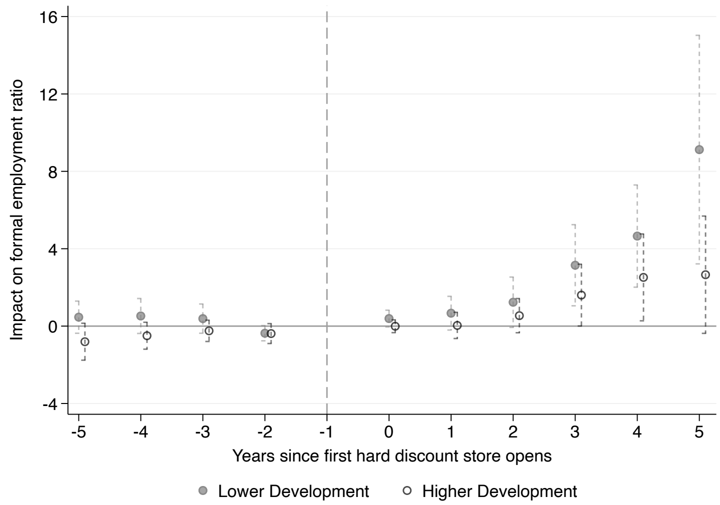
(b) Unemployment



(c) Inactivity

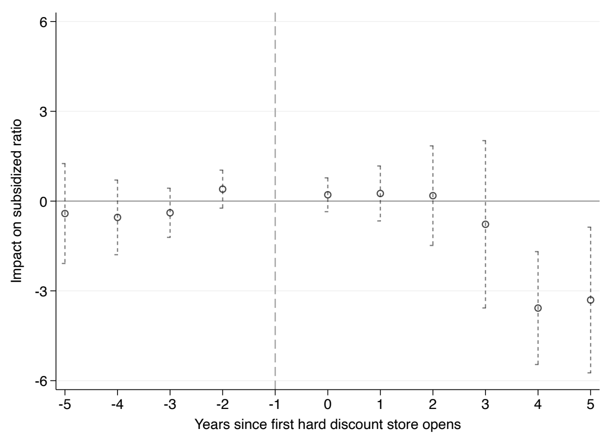
Note: We use the C&S and TWFE estimators. The dependent variables are yearly labor market rates. Regressions are weighted by local employment in 2010. The sample includes 191 treated municipalities. When outcomes are unbalanced, we restrict the analysis to pair-balanced observations observed in both pre- and post-treatment periods. Standard errors are clustered at the municipality level. 90% confidence intervals. Coefficients represent percentage point changes. Source: GEIH 2010–2018.

Figure F.4: Heterogeneous event study estimates on formal employment ratios



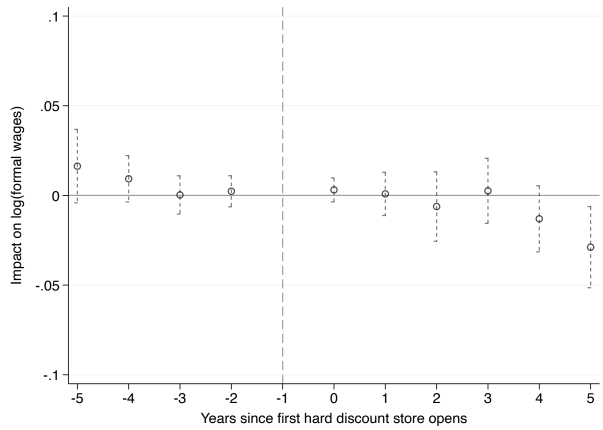
Note: We use the *C&S* estimator. The outcome is the formal employment ratio. We distinguish municipalities by baseline development: 191 municipalities with survey coverage are classified as higher development, while the remaining 181 municipalities without survey coverage are classified as lower development. Regressions are weighted using the local working-age population from the 2005 census. Standard errors are clustered at the municipality level. 90% confidence intervals. Source: PILA 2010–2018 in August.

Figure F.5: Event study estimates on subsidized social protection beneficiaries



Note: The dependent variable is the number of subsidized social-protection beneficiaries, multiplied by the share of individuals aged 15 to 59 in the 2018 census and divided by the working-age population in the 2005 census. Regressions are weighted by the local working-age population in the 2005 census. The number of treated municipalities is 372. Standard errors are clustered at the municipality level. 90% confidence intervals. The coefficients represent percentage point changes. Source: Health Ministry 2010-2018.

Figure F.6: Event study estimates on formal wages using administrative data



Note: We use the *C&S* estimator. The dependent variable is the logarithm of average wages. Regressions are weighted by the local working-age population in the 2005 census. The number of treated municipalities is 372. Standard errors are clustered at the municipality level. 90% confidence intervals. The coefficients represent log points. Source: PILA 2010-2018 in August.

F.2 Tables

Table F.1: Descriptive statistics for the estimation sample using administrative data

	Treated				Not yet treated			
	2011	**2013**	**2016**	**2018**	**2011**	**2013**	**2016**	**2018**
Proportion male workers	62.3%	61.6%	60%	61.4%	62.9%	62.1%	62.0%	66.4%
Average formal employment	6,781	4,609	5,572	3,856	2,146	2,554	2,115	1,318
	(6,079)	(6,412)	(8,397)	(6,139)	(4,090)	(5,011)	(4,101)	(3,437)
CHR workers (%)	5.1%	6.6%	9.3%	10.0%	6.5%	8.5%	9.3%	10.1%
Non-CHR workers (%)	94.8%	93.3%	90.6%	89.9%	93.4%	91.4%	90.6%	89.8 %
Employees (%)	64.0%	73.0%	75.7%	83.7%	63.9%	70.6%	74.2%	87.5%
Independent workers (%)	18.4%	18.4%	19.4%	12.7%	20.1%	22.5%	21.6%	9.4%
Min wage workers (%)	57.2%	53.6%	46.9%	53.4%	53.5%	54.8%	47.6%	54.4%
Average earnings	301.6	323.8	334.7	353.7	319.8	337.0	311.2	334.6
	(279)	(332)	(391)	(377)	(356)	(362)	(269)	(277)
Municipalities	10	53	151	291	362	319	221	81

Note: This table reports the mean of selected labor market indicators using administrative records from PILA by year and treatment status. A municipality is considered treated when the first hard discount store opens in the local market. The average total employment is computed for all treated municipalities that year and includes only full-time employees. "CHR workers" represent the proportion of formal full-time workers working in commerce, hotels, and restaurants. "Non-CHR workers" represent the proportion of formal full-time workers working in other industries outside commerce, hotels, and restaurants. "Employees" is the share of dependent workers in the total number of formal workers. "Independent workers" is the share of formal self-employed workers among total formal workers. "Min wage workers" is the share of formal full-time workers earning the minimum wage among total formal workers. In PILA, no workers earn less than the minimum wage, as by definition formal full-time workers cannot earn less than the monthly minimum wage. "Average earnings" is the average reported labor income of full-time formal workers. Standard deviations are in parentheses.

Table F.2: Correlation between municipal population growth and the year of hard discounter arrival

	Pop growth rate (%): 2010 vs 2005			2015 vs 2005				2020 vs 2010			
	(1) All	(2) PILA	(3) GEIH	(4) All	(5) All	(6) PILA	(7) GEIH	(8) All	(9) All	(10) PILA	(11) GEIH
HDS in municipality by 2013 (compared to later arrival)	-0.304	-1.131	-2.311	-0.959	-0.349	-0.109	-0.065	-0.670	-0.081	0.518	-0.237
	(0.836)	(0.996)	(1.452)	(1.916)	(0.450)	(0.587)	(1.100)	(2.392)	(1.246)	(1.602)	(2.336)
Population growth between 2005 and 2010 (%)					2.008***	2.021***	1.925***		1.934***	1.966***	1.785***
					(0.121)	(0.127)	(0.206)		(0.249)	(0.262)	(0.423)
Constant	4.901***	4.764***	5.431***	10.965***	1.123	1.152	1.537	15.540***	6.059***	6.124***	7.001**
	(0.348)	(0.371)	(0.550)	(0.714)	(0.716)	(0.742)	(1.349)	(0.791)	(1.471)	(1.529)	(2.779)
Observations	414	372	191	414	414	372	191	414	414	372	191
R-squared	.000311	.00354	.0134	.000701	.915	.915	.894	.000262	.652	.66	.599

Note: This table presents estimates of the correlation between municipal population growth and the year of hard discounter arrival. We regress three population growth rates on a dummy variable indicating whether hard discounters arrived in 2013 or earlier. Columns 1 to 3 use the rate of population growth between 2005 and 2010. Columns 4 to 7 use the rate of population growth between 2010 and 2015, and Columns 8 to 11 use the rate of population growth between 2010 and 2020. Columns 2, 6, and 10 include only the municipalities in our formal employment administrative registry sample. Similarly, Columns 3, 7, and 11 include only the municipalities in our sample with household survey data. Population estimates come from Colombia's official statistics agency, DANE. Robust standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table F.3: Labor income and working hours outcomes for estimation sample using survey data

	Treated				Not yet treated			
	2011	**2013**	**2016**	**2018**	**2011**	**2013**	**2016**	**2018**
Wages (USD)	274.7	300.3	305.4	300.4	276.4	294.3	274.6	282.7
	(110.9)	(60.6)	(85.8)	(77.1)	(69.5)	(95.9)	(61.4)	(70.5)
Wages: Informal sector (USD)	191.2	222.8	219.9	209.0	206.6	213.8	194.4	194.6
	(61.2)	(67.9)	(48.1)	(42.7)	(54.1)	(60.2)	(47.7)	(39.9)
Wages: Formal sector (USD)	372.4	413.3	417.7	441.7	434.6	453.1	443.9	479.8
	(114.2)	(74.6)	(106.8)	(133.5)	(110.2)	(129.4)	(114.3)	(107.9)
Working hours	47.1	45.3	46.1	45.7	47.0	46.4	45.9	43.8
	(1.9)	(3.7)	(2.5)	(3.0)	(3.2)	(3.0)	(3.0)	(3.7)
Working hours: Informal sector	44.6	43.8	44.2	44.1	46.2	45.5	45.5	42.2
	(3.0)	(4.4)	(3.7)	(3.8)	(3.6)	(3.9)	(3.9)	(3.6)
Working hours: Formal sector	51.7	50.2	49.3	48.4	49.4	49.2	47.7	46.3
	(1.4)	(2.6)	(2.2)	(3.1)	(4.3)	(3.6)	(4.7)	(5.0)
Municipalities	5	28	85	156	186	163	106	35
Average 2010 Employed Population	35,923	21,407	26,963	21,058	18,750	18,821	12,974	10,917

Note: This table shows the mean values of labor income and working hours indicators using the municipal panel of the GEIH, categorized by year and treatment status. The descriptive statistics are weighted by total employment in each municipality in 2010. Standard deviations are in parentheses.

Table F.4: Average *C&S* estimates of labor market rates

	(1) Employment rate	(2) Unemployment rate	(3) Inactivity rate
ATT_{pre}	0.718	-0.477	-0.439
	(0.536)	(0.398)	(0.443)
ATT_{post}	2.286***	-0.990	-1.886***
	(0.849)	(0.628)	(0.728)
$ATT_{post_{k=3,4,5}}$	3.242***	-1.758**	-2.369**
	(1.191)	(0.884)	(1.058)
N	1,719	1,674	1,713
Municipalities	191	191	191
Mean pre-treatment	70.3	11.3	20.9

Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: The dependent variable is the yearly labor market rate. Regressions were weighted by local employment in 2010. The number of treated municipalities is 191. Since the panel is not balanced for certain outcomes, we use only pair-balanced observations, that is, observations available in both pre- and post-treatment periods. Standard errors are clustered at the municipality level. The coefficients represent percentage point changes. Source: GEIH 2010-2018 in August.

Table F.5: Average formal employment effects by cohorts of treatment

	Admin	Survey
$g = 2011$	2.083**	2.853
	(0.864)	(2.173)
$g = 2012$	1.455*	1.705
	(0.768)	(1.300)
$g = 2013$	3.283	3.459
	(3.097)	(2.593)
$g = 2014$	1.417*	4.964***
	(0.834)	(1.098)
$g = 2015$	0.872	4.854*
	(1.234)	(2.735)
$g = 2016$	-0.111	0.408
	(0.485)	(1.485)
$g = 2017$	0.177	0.196
	(0.456)	(1.512)
$g = 2018$	-0.163	-0.401
	(0.316)	(2.412)

Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: We use the *C&S* estimator. The dependent variable in column (1) is formal employment using PILA, and in column (2) is formal employment using GEIH, both over the working-age population according to the 2005 census. Regressions were weighted with the local working-age population in 2005. Observed treated municipalities in PILA are 372, and in GEIH are 191. Standard errors are clustered at the municipality level. The coefficients represent percentage point changes. Source: PILA 2010-2018 in August, and GEIH 2010-2018.

Table F.6: Average *C&S* estimates of formal employment ratios by sector using administrative data

	(1) Total	(2) Retail	(3) CHR without retail	(4) Primary and secondary	(5) Services without CHR
ATT_{pre}	-0.417	-0.045*	0.030	-0.074	-0.100
	(0.294)	(0.025)	(0.090)	(0.205)	(0.211)
ATT_{post}	1.742***	0.108**	-0.290*	0.877***	0.670
	(0.608)	(0.044)	(0.158)	(0.296)	(0.474)
$ATT_{post_{k=3,4,5}}$	3.147***	0.170***	-0.397*	1.647***	1.013
	(1.049)	(0.066)	(0.228)	(0.426)	(0.756)
N	3,348	3,348	3,348	3,348	3,348
Municipalities	372	372	372	372	372
Mean pre-treatment	16	.5	1.7	4.5	12.5

Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: The dependent variable is formal employment over the working-age population according to the 2005 census. Regressions were weighted with the local working-age population in the 2005 census. The CHR refers to commerce, hotels, and restaurants. The primary and secondary sectors are manufacturing, construction, agriculture, and mining. Observed treated municipalities are 372. Standard errors are clustered at the municipality level. The coefficients represent percentage point changes. Source: PILA 2010-2018 in August.

Table F.7: Average *C&S* estimates of formal employment ratios by subsector using administrative data

	(1) Total	(2) Agriculture	(3) Manufacturing	(4) Construction
ATT_{pre}	-0.417	0.037	-0.210**	0.076
	(0.294)	(0.112)	(0.097)	(0.056)
ATT_{post}	1.742***	-0.005	0.611***	0.175**
	(0.608)	(0.160)	(0.167)	(0.082)
$ATT_{post_{k=3,4,5}}$	3.147***	0.028	1.118***	0.338***
	(1.049)	(0.226)	(0.268)	(0.122)
N	3,348	3,348	3,348	3,348
Municipalities	372	372	372	372
Mean pre-treatment	16	1.7	1.8	0.8

Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: The dependent variable is formal employment in each sector over the working-age population according to the 2005 census. Regressions were weighted by the local working-age population in the 2005 census. The number of treated municipalities is 372. Standard errors are clustered at the municipality level. The coefficients represent percentage point changes. Source: PILA 2010-2018.

Table F.8: Average *C&S* estimates of formal employment ratios by sector using survey data

	(1) Total	(2) Retail	(3) CHR without retail	(4) Primary and secondary	(5) Services without CHR
ATT_{pre}	0.215	-0.057	-0.066	0.190	0.148
	(0.738)	(0.182)	(0.158)	(0.365)	(0.538)
ATT_{post}	2.910***	0.554**	-0.069	1.911***	0.514
	(0.967)	(0.220)	(0.207)	(0.535)	(0.524)
$ATT_{post_{k=3,4,5}}$	4.501***	0.778***	0.196	2.724***	0.804
	(1.318)	(0.286)	(0.293)	(0.719)	(0.768)
N	1,719	1,719	1,719	1,719	1,719
Municipalities	191	191	191	191	191
Mean pre-treatment	28.3	2.6	2.2	8.4	15.1

Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: The dependent variable is formal employment in each sector relative to the working-age population according to the 2005 census. The CHR refers to commerce, hotels, and restaurants. The primary and secondary sectors are manufacturing, construction, agriculture, and mining. Regressions were weighted by the working-age population of the 2005 census. Observed treated municipalities are 191. Standard errors are clustered at the municipality level. The coefficients represent percentage point changes. Source: GEIH 2010-2018.

Table F.9: Average *C&S* estimates of formal employment ratios by subsector using survey data

	(1) Total	(2) Agriculture	(3) Manufacturing	(4) Construction
ATT_{pre}	0.215	-0.114	0.112	0.080
	(0.738)	(0.156)	(0.222)	(0.184)
ATT_{post}	2.910***	0.578*	0.920***	0.373*
	(0.967)	(0.348)	(0.288)	(0.194)
$ATT_{post_{k=3,4,5}}$	4.501***	0.766*	1.272***	0.581*
	(1.318)	(0.432)	(0.391)	(0.311)
N	1,719	1,719	1,719	1,719
Municipalities	191	191	191	191
Mean pre-treatment	28.3	2.1	4.5	1.3

Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: The dependent variable is formal employment in each sector over the working-age population according to the 2005 census. Regressions were weighted by the local working-age population in the 2005 census. Observed treated municipalities are 191. Standard errors are clustered at the municipality level. The coefficients represent percentage point changes. Source: GEIH 2010-2018.

Table F.10: Average *C&S* estimates of informal employment ratios by sector using survey data

	(1) Total	(2) Retail	(3) CHR without retail	(4) Primary and secondary	(5) Services without CHR
ATT_{pre}	0.980	0.085	0.413	-0.718	1.199***
	(0.782)	(0.357)	(0.318)	(0.473)	(0.417)
ATT_{post}	-1.105	0.281	-0.223	-0.806	-0.357
	(1.467)	(0.573)	(0.429)	(0.856)	(0.496)
$ATT_{post_{k=3,4,5}}$	-2.219	0.017	-0.674	-0.441	0.804
	(1.729)	(0.756)	(0.569)	(1.054)	(0.768)
N	1,719	1,719	1,719	1,719	1,719
Municipalities	191	191	191	191	191
Mean pre-treatment	45.3	9.7	7.6	12.6	15.5

Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: The dependent variable is informal employment in each sector over the working-age population according to the 2005 census. The CHR refers to commerce, hotels, and restaurants. The primary and secondary sectors are manufacturing, construction, agriculture, and mining. Regressions were weighted by local employment in 2010. Observed treated municipalities are 191. Standard errors are clustered at the municipality level. The coefficients represent percentage point changes. Source: GEIH 2010-2018.

Table F.11: Average *C&S* estimates of tax revenue ratios by type

	(1) All	(2) Non taxes	(3) Property	(4) Industry and commerce	(5) Gasoline	(6) Other taxes
ATT_{pre}	2.321	1.696	-2.196*	0.718	0.122	3.678*
	(4.003)	(1.470)	(1.232)	(1.927)	(0.311)	(2.158)
ATT_{post}	10.138**	2.265	3.599*	6.291**	0.059	0.189
	(4.740)	(1.735)	(2.025)	(2.939)	(0.509)	(1.738)
$ATT_{post_{k=3,4,5}}$	14.091*	3.886	5.822*	10.967**	0.270	-2.968
	(7.374)	(2.781)	(3.161)	(4.741)	(0.718)	(2.582)
N	3,339	3,339	3,339	3,339	3,339	3,339
Municipalities	371	371	371	371	371	371
Mean pre-treatment	134.3	14.2	42.2	44.3	13.8	34.0

Standard errors are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Note: The dependent variable is the type of tax or revenue over all revenues in 2010. Regressions were weighted by the municipality's 2010 revenues. Observed treated municipalities are 371. Standard errors are clustered at the municipality level. The coefficients represent percentage point changes. Source: DNP, 2010-2018.